Article

CLOP-DiT: Structured-Metadata-Conditioned Single-Cell Latent Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

Zeyu Fu [1,*]

[1] State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing 400038, China; zeyu.fu@tmmu.edu.cn

Simple Summary

CLOP-DiT is a computational pipeline that generates synthetic single-cell gene expression profiles from structured biological descriptions (cell type, tissue, organism, marker genes, and disease context). It first learns to align text descriptions with real cell data in a shared mathematical space, then uses a diffusion model to generate new cell states matching a given description. The generated cells capture correct cell-type identity and marker gene patterns, but do not yet reproduce the full cell-to-cell variability seen in real single-cell experiments. This work demonstrates that text-guided single-cell generation is feasible as a proof of concept, opening directions for future simulation and hypothesis-generation tools in biology.

Abstract

We present CLOP-DiT, a pipeline that combines Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained by flow matching to generate single-cell latent states from structured biological metadata. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings in a shared 512-dimensional space; DiT then samples latents conditioned on a five-field template (cell type, tissue, organism, marker genes, disease). Across 69 cell types from 80 GEO datasets (220,304 cells), CLOP-DiT achieves 36.9% KNN accuracy and 81.0% steering (CFG = 2.0), while a high-diversity regime reaches diversity ratio 0.93 at 80.7% steering (CFG = 1.0). A frozen scGPT decoder maps generated latents to gene expression with high mean concordance (Pearson $r > 0.999$), but per-gene variance correlation is near zero, indicating loss of cell-to-cell heterogeneity. Conditioning-field ablation confirms marker genes are the dominant steering signal, reducing accuracy from 99.8% to 62.4% upon removal. External validation against PanglaoDB and CellMarker 2.0 supports biological plausibility. However, generated cells remain distinguishable from real data (discriminator AUC 0.656), a Gaussian baseline outperforms CLOP-DiT on common metrics, and a pilot rare-cell augmentation study was negative. CLOP-DiT establishes the feasibility of structured-metadata-conditioned single-cell latent generation but not yet its practical utility for downstream biology.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; structured-metadata conditioning; flow matching; latent generation; foundation model

Received:
Revised:
Accepted:
Published:

 https://doi.org/10.3390/biology1010000

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed the study of cellular heterogeneity by enabling transcriptome-scale profiling at single-cell resolution [1,2]. Yet generating synthetic single-cell states from structured biological descriptors remains difficult. A model that could condition on cell identity, tissue context, organism, marker genes, and disease state would be attractive for controlled simulation and hypothesis generation, but it should be judged by whether it preserves biologically meaningful structure rather than by the fluency of the textual prompt alone.

Most current single-cell generative models condition on categorical labels or perturbation metadata rather than richer multi-field descriptions. Variational autoencoders such as scVI [3,4] and scGen [5], geometric generative models [6], and foundation models such as Geneformer [7] and scBERT [8–11] broaden the modeling landscape, but they do not jointly learn a structured text–cell alignment space and a conditional diffusion-style generator. Cell2Sentence [12] and GenePT [13] move closer to language-based biology, yet they address different problem formulations and do not combine contrastive cross-modal alignment with latent flow matching. Table 1 summarizes the key differences.

Table 1. Comparison of CLOP-DiT with related methods for conditional single-cell generation and annotation. "Cond. modality" indicates the conditioning input type; "Gen. paradigm" indicates the generation approach.

Method	Cond. Modality	Gen. Paradigm	Output
scVI [3]	Categorical (batch/type)	cVAE	Expression
scGen [5]	Categorical (perturbation)	VAE + shift	Expression
Geneformer [7]	Rank-value tokens	Masked LM	Cell embeddings
Cell2Sentence [12]	Gene-rank text	LLM decoding	Gene-rank text
GenePT [13]	Gene text embeddings	None (annotation)	Annotations
CLOP-DiT (ours)	**Structured 5-field text**	**DiT + flow matching**	**Gene expression**

Recent advances in contrastive multimodal learning and diffusion modeling motivate a direct bridge between biological descriptions and cell representations. CLIP/SigLIP-style objectives show that strong cross-modal alignment can emerge from contrastive training [14,15], while Diffusion Transformers and flow matching provide scalable conditional generation in continuous latent spaces [16–19]. These ingredients suggest a natural design for structured biological conditioning: first build a well-separated text–cell condition space, then learn a conditional sampler in a biologically meaningful latent representation.

We present CLOP-DiT, a two-stage pipeline that aligns structured metadata with scGPT cell embeddings and then samples scGPT latents conditioned on that aligned representation. The input is a fixed five-field template—cell type, tissue, organism, marker genes, and disease context—so the method should be interpreted as structured-metadata conditioning rather than free-form language understanding. A frozen scGPT decoder can optionally map generated latents back to expression, but because this decoder is many-to-one, latent-space metrics are the primary indicators of generative quality.

The main contributions are fourfold. First, CLOP uses PrototypeSigLIP with ZCA-whitened BiomedBERT embeddings to align text and cell representations in a shared 512-dimensional space and substantially improve inter-type separability. Second, a 1D Diffusion Transformer performs conditional flow matching in that latent space to generate type-specific cell embeddings. Third, we evaluate the model on 69 deduplicated cell types from 80 GEO datasets using complementary latent, diversity, and downstream biological metrics. Fourth, we report both the strengths and the limits of the approach: structured conditioning produces clear above-chance type specificity, but variance preservation, covariance preservation, and rare-cell augmentation remain unresolved.

Across the main operating points, CLOP-DiT reaches 36.9% KNN accuracy and 81.0% steering in a high-fidelity regime, while a high-diversity regime achieves a diversity ratio of 0.93 at 80.7% steering. Multi-seed replication shows stable steering and diversity, whereas decoder-level analyses reveal strong mean reconstruction but weak preservation of within-population heterogeneity. CLOP-DiT should therefore be viewed as a proof-of-concept structured-metadata-to-latent conditional sampler with a clear roadmap for improvement, rather than as a finished tool for downstream biological analysis.

2. Materials and Methods

Figure 1 provides an overview of the pipeline. We organize the Materials and Methods in three parts: (1) Dataset curation and preprocessing (2.1); (2) Model architecture and training—covering CLOP alignment (2.2), DiT generation (2.3), and scGPT decoding (2.4); (3) Evaluation framework and metrics (2.5). Together, these components form a reproducible pipeline for text-conditioned single-cell generation.

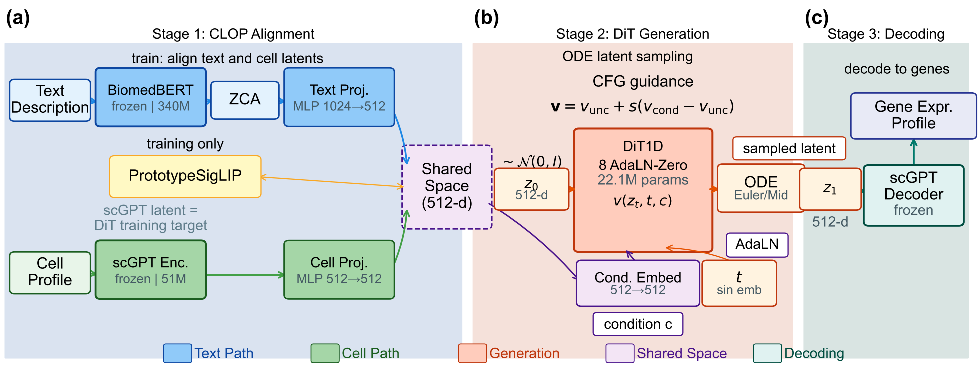

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)** aligns structured text descriptions and scGPT cell embeddings in a shared 512-dimensional space after BiomedBERT whitening and projection. **Stage 2 (DiT)** performs conditional flow matching in scGPT latent space and generates a target latent from Gaussian noise using classifier-free guidance at inference. **Stage 3 (Decoding)** maps the generated latent back to gene expression through the frozen scGPT decoder.

2.1. Dataset

We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection [20,21] (2,000 highly variable genes per dataset, selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured text caption (not free-form natural language) incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template:

```
{cell type}, tissue: {tissue}, organism: {organism},
markers: {top 5 DE marker genes}, context: {disease/condition}
```

where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test [22] on the training set. As an independent check on marker gene circularity, we validated these training-set-derived markers against PanglaoDB [23] and CellMarker 2.0 [24]; the high Jaccard overlap (0.39–0.45) indicates that the selected markers reflect biological consensus (see Discussion). The exact template and marker gene sources are available in `src/data_pipeline/subcluster_annotation.py` (caption generation) and `scripts/data_prep/02b_enrich_descriptions.py` (evidence enrichment).

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits, and were chosen to span the major tissue and cancer categories in the corpus (lung, gastric, skin, liver, blood) via stratified selection across tissue types; the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2. Within the 72 training datasets, a cell-type-stratified split reserves $\sim$10% of cells per type for training-time validation, ensuring that all 69 deduplicated cell types are represented in both partitions. This held-out design tests interpolation within the training distribution; the 8 held-out datasets share tissue types and cell populations with the training corpus and do not represent a stringent out-of-distribution generalization test.

The relationship between the 1,088 text groups and the 69 evaluated cell types is as follows. The 1,088 text groups correspond to sub-cluster-level descriptions: each Leiden [25] cluster within each dataset receives a unique text caption. Many sub-clusters share the same canonical cell type (e.g., "CD8+ T cells" in lung and liver datasets produce separate text groups). For evaluation, text groups are consolidated by core cell-type identity via a deduplication pipeline that (a) removes uncharacterized or ambiguous sub-clusters (mapped to null in the annotation pipeline) and (b) merges cross-dataset sub-clusters into canonical types, yielding 69 evaluable cell types after deduplication. This consolidation is implemented in `scripts/analysis/deduplicate_captions.py` and produces the deduplicated cache (`data/cached_latents_v5.2/METADATA_DEDUP.json`: 1,088 original captions $\rightarrow$ 69 deduplicated types, 167,245 cells). KNN accuracy is computed over these 69 consolidated types (random chance = $1/69 \approx 1.45\%$). All headline generation metrics, bootstrap confidence intervals, and benchmark comparisons operate on this consistent 69-type evaluation set.

The 80 datasets comprise both *Homo sapiens* (59 datasets) and *Mus musculus* (21 datasets) [26], predominantly from tumor microenvironment and developmental contexts. Consequently, all generation and evaluation results should be interpreted as applying to human and mouse cancer and developmental single-cell biology; generalization to other organisms, non-cancer tissues, or perturbation conditions has not been tested.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12.0), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop.yaml`.

Hardware and compute budget.

All training and evaluation were conducted on a single NVIDIA GeForce RTX 5090 Laptop GPU (24 GB VRAM) using PyTorch 2.1 with CUDA 12.0 and automatic mixed precision (AMP/FP16). The modest compute requirements (<2 GPU-hours total, <24 GB VRAM peak) mean that any Ampere-or-later GPU with $\geq$16 GB VRAM should suffice for reproduction. CLOP alignment training (60 epochs, batch size 1024) completes in approximately 10 minutes ($\sim$10 s/epoch). DiT flow-matching training (200 epochs, batch size 1024) completes in approximately 75 minutes ($\sim$23 s/epoch). End-to-end figure reproduction from cached embeddings (`scripts/pipeline/regenerate_report.sh`) takes under 30 minutes. Total compute for a single full training run (CLOP + DiT + evaluation + visualization) is under 2 GPU-hours; no multi-GPU or distributed training was used. Inference for 69 cell types ($\sim$6,900 generated cells) takes approximately 2 minutes with 10-step Euler integration.

2.2. CLOP: Contrastive Language–Omics Pretraining

CLOP aligns structured metadata descriptions and cell profiles into a shared 512-dimensional embedding space using a PrototypeSigLIP contrastive loss, producing the well-separated condition space required by the DiT. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: 1024 → 1024 → 1024 → 512; cell: 512 → 512 → 512 → 512) with BatchNorm [27], GELU activation [28], and dropout [29] (0.2) project both modalities onto the L_2-normalized unit hypersphere.

ZCA whitening.

Prior to projection, Zero-phase Component Analysis (ZCA) whitening is applied to the BiomedBERT embeddings to decorrelate dimensions and mitigate the collapse typical of frozen BERT representations. The transform is fitted once on the training text embeddings using the full 1024×1024 covariance matrix with regularisation $\epsilon = 10^{-4}$, then reused unchanged for validation and inference text. All components are retained; the goal is decorrelation rather than dimensionality reduction. In preliminary sensitivity checks, ZCA produced substantially lower pairwise cosine similarity than mean-centering or LayerNorm, and it was therefore adopted as the production preprocessing step.

The CLOP loss is:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{TC}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{CT}^\top}{\tau}, \text{arange}(B)\right)\right] \tag{1}$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B \times 512}$ are L_2-normalized projection matrices (each row is a projected text or cell embedding, respectively), B is the batch size, $\mathbf{TC}^\top \in \mathbb{R}^{B \times B}$ is the pairwise cosine similarity matrix, τ is a logit-scale parameter (following the CLIP convention where τ multiplies rather than divides the similarity matrix), and label smoothing $\epsilon = 0.1$ is applied. The targets $\text{arange}(B) = [0, 1, \ldots, B-1]$ enforce identity matching: each text embedding should match its corresponding cell embedding in the batch. In the production configuration, $\tau = 14.0$ is **fixed** (effective softmax temperature $1/\tau \approx 0.071$); the value was selected via grid search over $\{7.0, 10.0, 14.0, 20.0\}$ to maximise inter-type cosine separation on the validation split, yielding the sharpest condition space before training-accuracy collapse. The ablation study (Table 5) was conducted on a prior baseline (v8.2) that used a *learnable* temperature; the "Fixed temperature" ablation row therein shows that switching from learned to fixed temperature reduced prototype accuracy by −4.61 pp under that earlier configuration. The production model (v9.3) subsequently adopted fixed temperature alongside a stratified validation split and updated regularization schedule, under which the fixed-temperature setting achieved satisfactory downstream generation quality (Section 3). The loss function follows the PrototypeSigLIP formulation, which aligns text prototypes to cell-group centroids rather than individual cells, with a cohesion regularization weight of 0.15. The CLOP aligner has 3.02M trainable parameters and is trained for 60 epochs with AdamW [30] (lr $= 5 \times 10^{-4}$, weight decay 0.06, cosine schedule [31] with 5-epoch warmup).

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of ∼130×. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 8-head self-attention (head dimension 64) and feed-forward layers (512 → 2048 → 512). Timestep information is encoded via sinusoidal embeddings projected to 512 dimensions, and CLOP condition vectors are projected from 512 to 512 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ (element-wise summation, not concatenation) modulates each block through 6 AdaLN-Zero parameters ($\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$), all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [17,32]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \tag{2}$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution (μ=0, σ=1), which concentrates training on intermediate timesteps where the velocity field is most complex [33]. During training, classifier-free guidance (CFG) [34] is enabled by dropping the condition with 15% probability; when the condition is dropped, c_{emb} is replaced by a zero vector of the same dimension ($\mathbf{0} \in \mathbb{R}^{512}$), serving as the unconditional null token. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \tag{3}$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA [35] (decay = 0.9999).

2.4. Decoding via scGPT

Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are mapped back to gene expression using the frozen scGPT decoder. This step is useful for downstream biological inspection, but it introduces an important limitation: the decoder is many-to-one, so diverse latent vectors near the training manifold can decode to similar expression profiles. We therefore treat latent-space metrics—KNN accuracy, steering, diversity ratio, and centroid cosine—as the primary indicators of generative quality, whereas expression-space analyses should be interpreted as decoder-induced compatibility checks rather than direct measures of generation fidelity. Breaking this bottleneck will require a trainable decoder or lightweight adapter, which is left to future work.

2.5. Evaluation Framework

We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. PCA is fitted on the real training split only and applied identically to generated embeddings; the same PCA transform is used across all method comparisons. Random chance is $1/69 \approx 1.45\%$. Because real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, small embedding perturbations can substantially affect KNN accuracy; the biological meaning of absolute KNN values should therefore be interpreted cautiously.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%. **Note:** At intermediate-to-high CFG values (e.g., CFG $\geq$ 1.5) with multi-step solvers, steering accuracy can saturate at 1.0 with bootstrap CI [1.0, 1.0], indicating a ceiling effect; this metric loses discriminative power once guidance is sufficient.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values $<$1 indicate mode sharpening; values $>$1 indicate excess noise.
4. **Coverage and density**: Coverage measures the fraction of real samples that have at least one generated neighbor within the real data's k-NN radius (k=5); it quantifies

mode coverage. Density measures the average number of generated samples falling within each real sample's k-NN radius, normalised by k; it quantifies precision. Both are computed in PCA-50 embedding space.

5. **Centroid cosine similarity**: The cosine similarity between the real and generated cell-type centroids, computed in the 512-dimensional scGPT embedding space (uncentered; $\cos(\mu_{\mathrm{real}}, \mu_{\mathrm{gen}})$). Reported per-type and averaged across 69 types.
6. **Downstream biological concordance**: Clustering alignment (ARI [36], NMI [37]), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 16d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

Caveat on Fréchet Distance (FD). FD [38] compares distributional moments (mean and covariance) between real and generated samples. While widely used for unconditional evaluation, FD can be misleading for *conditional* generation: a model that reproduces the global marginal distribution but ignores conditioning achieves a low FD without having learned the target conditional. FD is therefore reported as a secondary metric throughout; the primary metrics (KNN accuracy, steering, diversity ratio) are specifically designed for conditional evaluation.

The biological evidence is organized by figure family: Figures 8–10 establish marker-level and gene-level fidelity; Figures 12–13 and 14 provide robustness readouts for guidance and diversity; Figures 15–16 report benchmarking; Figures 17 and 18 test downstream biology. Figure 2 provides a schematic overview of the complete evaluation pipeline, including data splitting, PCA fitting, KNN training, and metric aggregation stages.

Having established the methodology and evaluation framework, we now present the empirical results.

2.6. Statistical Analysis

Primary endpoints. Two operating points were pre-specified as primary comparisons: CFG = 2.0 with 10-step Euler integration (high-fidelity regime, selected because KNN accuracy plateaus at CFG 2.0–3.0 in Table A2; the lower end was chosen to minimize diversity loss) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime, selected to match DivR $\approx$ 1.0). The primary metrics are KNN top-1 classification accuracy (69-class, k=15, cosine metric on PCA-50 features) and steering accuracy (pairwise directional test, 1000 random pairs).

Exploratory analyses. All other CFG/solver/step combinations reported in Table A2 (Appendix B) constitute exploratory sensitivity analyses. No correction for multiple comparisons (e.g., Bonferroni, Holm) was applied to these comparisons. Rankings and performance claims refer only to the two primary operating points unless explicitly stated otherwise.

Uncertainty quantification. Bootstrap 95% confidence intervals (percentile method, B=1000 resamples, seed 42) are reported for all headline metrics where per-unit (per-cell or per-type) data are available (see Results). These CIs capture evaluation sampling variability but not training-run variance, as all results derive from a single training run (seed 42).

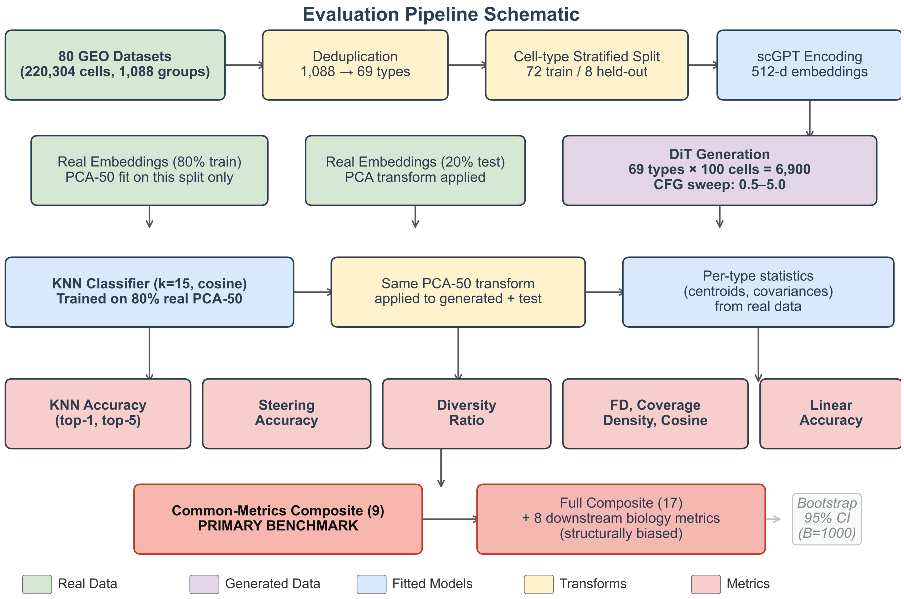

Figure 2. Evaluation pipeline schematic. Data flow from raw datasets through deduplication, stratified splitting, scGPT encoding, PCA fitting (on real training split only), KNN classifier training, and metric computation. The common-metrics-only composite (9 shared distributional metrics, marked in red) is the primary benchmark; the full 17-metric composite includes 8 downstream biology metrics reported only by CLOP-DiT and is structurally biased. Bootstrap confidence intervals (B=1,000) are computed by resampling the 69 evaluation types.

Formal superiority tests (e.g., permutation tests) were not conducted; all comparisons are descriptive.

Multi-seed replication. To characterise inter-run variance, we trained three independent CLOP + DiT pipeline replicas (seeds 42, 123, 456) with identical hyperparameters and report mean $\pm$ standard deviation for all core metrics in Table 3 (see Section 3.4).

Limitations of the statistical design. (1) No formal pre-registration: operating points were selected based on interpretable criteria post-hoc rather than by pre-registered protocol; this study is therefore exploratory rather than confirmatory. (2) No multiple-comparison correction for the exploratory CFG sweep.

2.7. Composite Score Formulation

The composite benchmark score aggregates all active metrics into a single ranking index. For each metric m with direction $d_m \in \{\text{lower}, \text{higher}\}$, let $v_{m,j}$ denote the raw value for method j. We compute a min–max normalised score:

$$s_{m,j} = \begin{cases} 1 - \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{lower}, \\ \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{higher}, \end{cases} \tag{4}$$

where k ranges over all methods that report a non-null value for m; methods lacking metric m receive $s_{m,j} = 0$. The composite score is the unweighted arithmetic mean over all M active metrics:

$$C_j = \frac{1}{M} \sum_{m=1}^{M} s_{m,j}. \tag{5}$$

Because not all methods support downstream biology (Section 2.5, axis 6), they receive $s = 0$ on those metrics, which inflates the score of methods that do. In the current benchmark, 8 of 17 active metrics are reported only by CLOP-DiT; the remaining 9 distributional metrics are shared by all methods. **To enable fair like-for-like comparison, the common-metrics-only composite ($C_j^{\textbf{common}}$, restricted to the 9 shared distributional metrics) serves as the primary benchmark figure.** The full composite (C_j over all $M = 17$ metrics) is reported as a supplementary capability ranking but should not be used for method comparison, as the 8 exclusive metrics structurally bias it in favour of CLOP-DiT. Table 2 and the per-metric panels in Figure 16 report individual metrics for transparent comparison. Bootstrap 95% confidence intervals (percentile method, B=1,000) for all headline generation metrics are provided in the supplementary materials (Table S5).

3. Results

We report in-distribution results along the four evaluation axes defined in the Methods, organized by evidence type: training dynamics and embedding space (Figures 3–4); core generation quality metrics (Figures 5–6 and 7); gene-level fidelity (Figures 8–10); conditioning robustness and diversity trade-offs (Figures 11–13 and 14); baseline comparisons and benchmarking (Figures 15–16); and downstream biological validation (Figures 17–18). This structure allows readers to assess evidence quality at multiple resolutions, from training convergence through partial downstream biological utility.

3.1. Training Dynamics

Figure 3 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 60, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%)—a 256-way contrastive ranking task in which the model must match each text to its corresponding cell from a full mini-batch—and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a final validation velocity cosine of 0.990 with EMA, indicating near-perfect velocity field prediction. Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 → 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

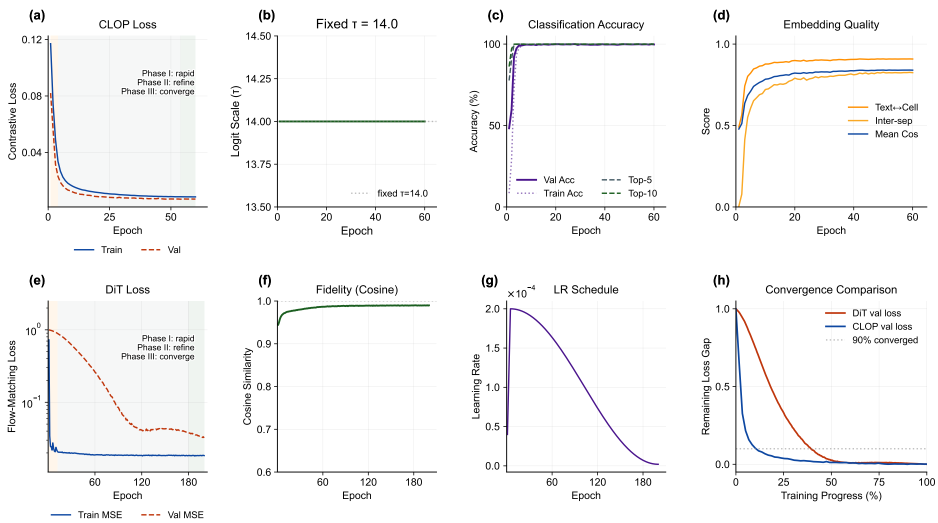

Figure 3. Training dynamics of the CLOP-DiT pipeline. (**a–d**) CLOP loss, fixed logit scale, batch accuracy, and embedding-quality metrics over 60 epochs. (**e–h**) DiT train/validation loss, velocity cosine similarity, learning-rate schedule, and normalized convergence comparison across training progress. Together, the panels show stable CLOP separation and near-converged DiT velocity prediction.

3.2. CLOP Embedding Space

Figure 4 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP [39]: text and cell embeddings jointly projected into the shared 512-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

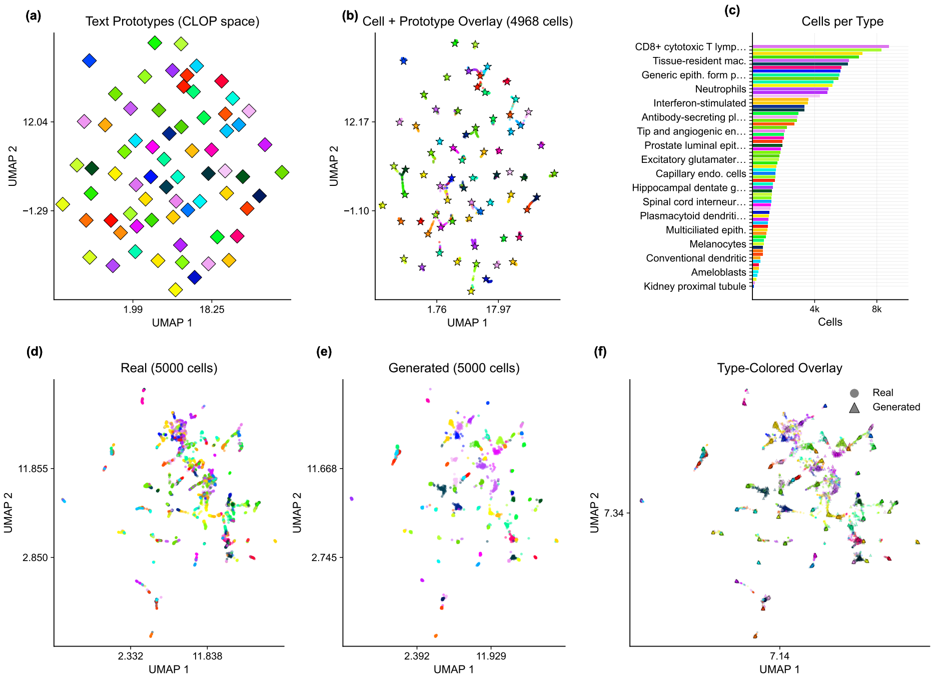

Figure 4. Embedding space analysis. (**a–c**) CLOP-projected text and cell embeddings form aligned clusters in the shared space. (**d–f**) Real and generated scGPT embeddings occupy overlapping but distinguishable regions, indicating that conditional generation follows the coarse structure of the data manifold without fully matching real-cell geometry.

3.3. Core Evaluation Metrics

Table 2 and Figure 5 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (25× random chance of 1.45%; 52 points below the 89% real-data baseline), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Note that the Fréchet distance (FD) reported in Table 2 is often misleading for conditional generation, because global mean/covariance matching tends to favour unconditional baselines; FD should therefore be interpreted alongside the directional metrics (KNN, steering, diversity). Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering = 47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Bootstrap confidence intervals. To quantify evaluation sampling uncertainty, we computed bootstrap 95% CIs by resampling the 69 cell types in the generation cache with replacement (B=1,000, percentile method). All 69 deduplicated types have both real and generated cells, so the bootstrap covers the full evaluation set consistently. At an intermediate reference configuration (CFG = 1.5, Midpoint, 20 steps; selected to balance

Table 2. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 *k*-nearest-neighbor accuracy among 69 deduplicated cell types (random = 1.45%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy in embedding space (trained on real, evaluated on generated; distinct from the expression-space downstream classifier in Figure 17). FD: Fréchet distance [38] (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

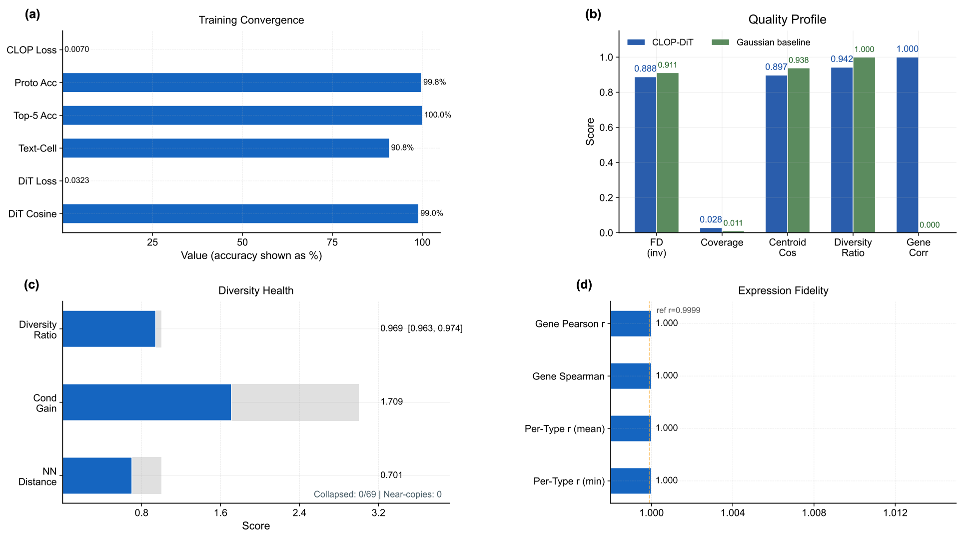

Figure 5. Metrics dashboard summarizing conditional generation quality. (**a**) Training convergence summary. (**b**) Multi-metric comparison between CLOP-DiT and the Gaussian baseline, including inverse FD, coverage, centroid cosine, diversity ratio, and per-gene Pearson correlation. (**c**) Diversity health gauges. (**d**) Expression-fidelity summary across configurations.

fidelity and diversity between the two primary operating points): KNN top-1 = 0.509 [0.421, 0.586], KNN top-5 = 0.728 [0.658, 0.794], steering = 1.000 [1.000, 1.000] (**ceiling effect**: all bootstrap resamples achieve perfect pairwise accuracy at this configuration, indicating that the steering metric saturates at moderate-to-high CFG and becomes non-discriminative; this ceiling should be considered when interpreting steering differences across configurations), diversity ratio = 0.969 [0.963, 0.975], linear accuracy = 0.583 [0.518, 0.646], and centroid cosine = 0.898 [0.875, 0.913]. The narrow diversity ratio and centroid cosine CIs reflect uniformly high performance across types; the wider KNN CIs reflect type-dependent classification difficulty. These CIs capture evaluation sampling variability; inter-run variance across three training seeds is reported separately in Table 3.

3.4. Multi-Seed Validation

To quantify inter-run variance from stochastic initialization, data shuffling, and mini-batch ordering, we trained three independent CLOP + DiT pipelines (seeds 42, 123, 456) with identical hyperparameters and evaluated each under both operating regimes (Table 3).

Table 3. Multi-seed validation (3 independent training runs). Mean $\pm$ standard deviation for core metrics under both operating regimes. All three seeds use identical hyperparameters (Table A3), identical data splits, and identical evaluation protocol.

	High-Fidelity (CFG=2.0, Euler)		**High-Diversity (CFG=1.0, Midpoint)**	
Metric	**Mean $\pm$ Std**	**Range**	**Mean $\pm$ Std**	**Range**
KNN Top-1	0.348 ± 0.082	[0.258, 0.419]	0.273 ± 0.048	[0.220, 0.312]
KNN Top-5	0.535 ± 0.016	[0.524, 0.553]	0.465 ± 0.038	[0.430, 0.504]
Steering	0.808 ± 0.009	[0.798, 0.817]	0.801 ± 0.008	[0.792, 0.804]
Diversity Ratio	0.511 ± 0.015	[0.496, 0.525]	0.924 ± 0.016	[0.906, 0.937]
Linear Acc.	0.488 ± 0.023	[0.465, 0.511]	0.355 ± 0.097	[0.257, 0.452]
Centroid Cos.	0.892 ± 0.012	[0.878, 0.900]	0.892 ± 0.010	[0.880, 0.898]

Inter-run standard deviations are small relative to effect sizes: steering accuracy varies by < 1 percentage point ($\sigma = 0.009$), diversity ratio by < 2 percentage points ($\sigma = 0.015$–0.016), and centroid cosine by $\sim 1\%$ ($\sigma = 0.010$–0.012). KNN top-1 accuracy exhibits the largest inter-seed variance ($\sigma = 0.082$ in the high-fidelity regime), reflecting the sensitivity of nearest-neighbor classification to the decision boundary in the tightly-packed scGPT embedding space. Critically, the qualitative conclusions—25$\times$ above random chance, $\sim$52-point gap from real data, near-ideal diversity at CFG = 1.0—are stable across all three seeds.

3.5. Per-Type Generation Quality and Text–Cell Alignment

Beyond aggregate metrics, per-type analysis reveals which cell types benefit most from text conditioning and where failures concentrate. Figure 6 presents per-type generation quality. The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across all 69 deduplicated cell types; the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as color. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. Figure 7 shows the text–cell alignment analysis, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

3.6. Marker Gene Fidelity

A key question for conditional generation is whether cell-type-specific marker gene signatures are preserved. Figure 8 compares the expression of canonical marker genes between real and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific

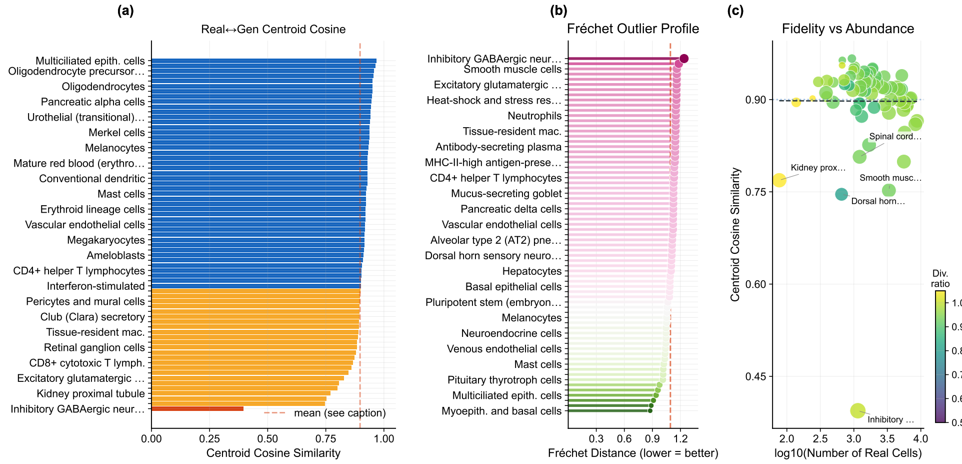

Figure 6. Per-type generation fidelity. (**a**) Per-type centroid cosine similarity ranked across cell types, (**b**) Fréchet outlier profile with the global mean marked, and (**c**) abundance–fidelity scatter (bubble size ∝ Fréchet distance, color encodes diversity ratio). Cell-type names are mapped by rank position in the supplementary materials.

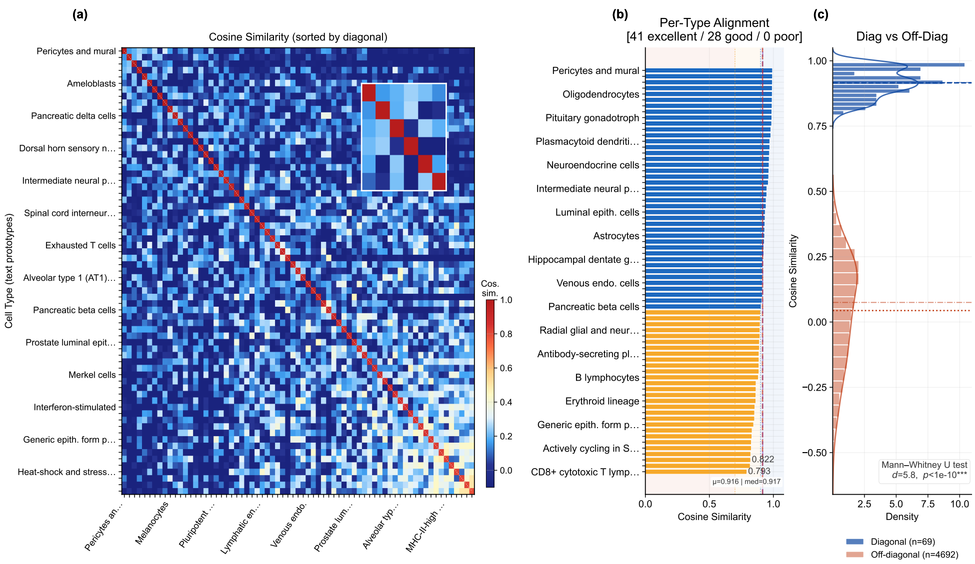

Figure 7. Text–cell alignment. (**a**) 69×69 cosine similarity heatmap between text and cell centroids, sorted by diagonal similarity. Mean diagonal similarity $\mu_{\text{diag}} = 0.916$ ($\sigma = 0.053$), mean off-diagonal $\mu_{\text{off}} = 0.044$ ($\sigma = 0.205$); Cohen's $d = 5.8$, separation ratio 20.9×, Mann–Whitney U $p < 10^{-10}$. (**b**) Per-type alignment bars color-coded by quality tier (41 excellent ≥ 0.9, 28 good, 0 poor). (**c**) Diagonal vs. off-diagonal distribution comparison with KDE overlay; the two populations are well separated ($\Delta = 0.872$, bootstrap 95% CI [0.875, 0.913]).

expression signatures. This directly shows that the biological signal resides in the intended marker genes rather than only in latent-space summary statistics.

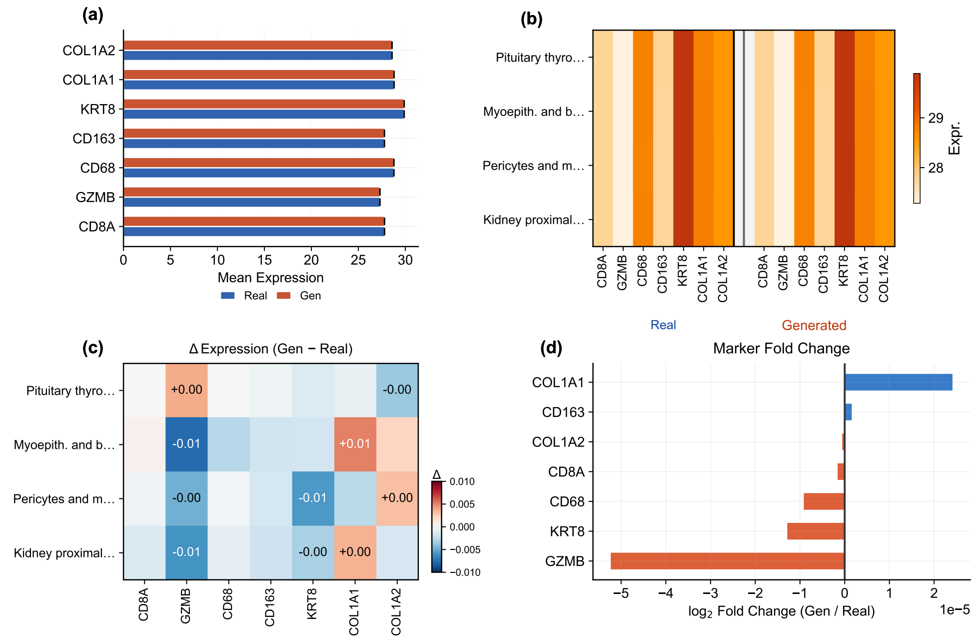

Figure 8. Marker gene comparison. (**a**) Paired bar chart of canonical marker gene expression in real (solid) vs. generated (hatched) cells, with bar colors grouping markers by lineage family (CD8 T, myeloid, epithelial, stromal); fold-change annotations flag markers deviating from unity. (**b**) Per-type $\times$ marker dual heatmap showing real and generated expression side by side for selected cell types. (**c**) Difference heatmap (Δ = generated $-$ real) with cell annotations for large deviations. (**d**) Fold-change waterfall for all markers, with a shaded band marking the $\pm 5\%$ agreement zone. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

3.7. *Expression Correlation and Dimension Analysis*

To assess gene-level fidelity beyond marker genes, Figures 9 and 10 quantify the per-gene Pearson correlation between real and generated mean expression across cell types and provide a broader view of expression distribution properties, including norm matching (generated mean norm 20.97 vs. real 20.89) and per-dimension statistics. Together they localize biological meaning in gene-wise means, variances, and marker-level residuals rather than only in embedding geometry.

3.8. *Conditioning Landscape and Diversity Trade-Offs*

Figure 11 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance. Together with Figures 12, 13, and 14, this section is the main robustness analysis for whether control can be increased without erasing biologically meaningful heterogeneity.

Building on this conditioning structure, a sweep over eight CFG settings (0.5 to 5.0) with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 12). KNN accuracy saturates near CFG = 2.0–3.0 (41$\times$ random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.930) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

Figure 13 and Figure 14 further quantify the two operating regimes: high-fidelity (CFG ≥ 2.0, DivR ≈ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augment the global trade-off curves with per-type diversity-tail diagnostics. Expression-level diversity measurements confirm that the Midpoint solver best preserves gene-level variance.

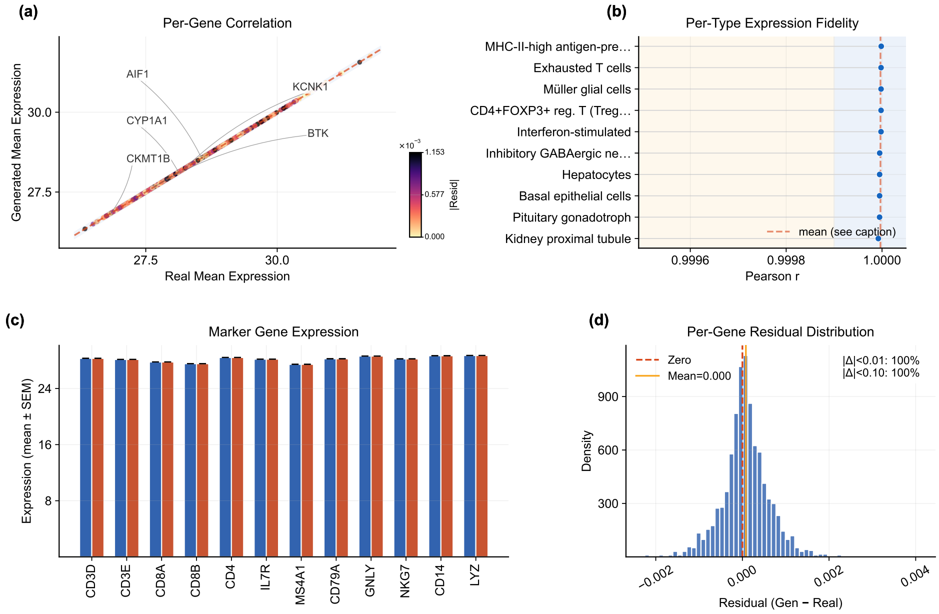

Figure 9. Expression correlation analysis. (**a**) Per-gene mean expression scatter (real vs. generated), with point color encoding absolute residual magnitude and outlier genes annotated. (**b**) Per-type Pearson r lollipop chart with threshold-band shading. (**c**) Marker-gene expression comparison with error whiskers. (**d**) Per-gene residual distribution (generated − real) with percentages within tolerance bands.

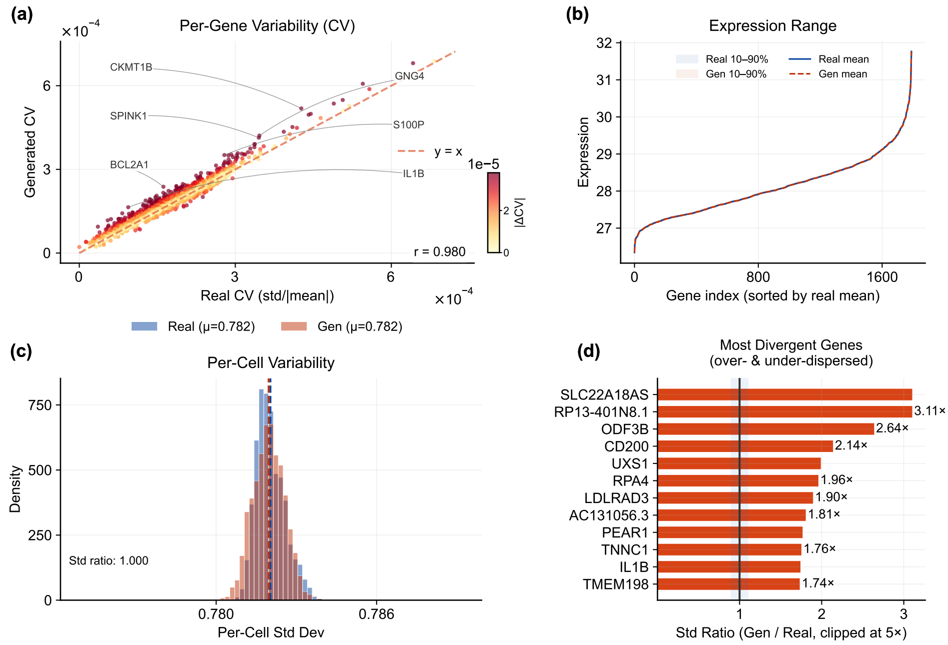

Figure 10. Expression distribution analysis. (**a**) Per-gene coefficient-of-variation scatter (real vs. generated) with annotated outliers. (**b**) Expression-range ribbons for real and generated cells. (**c**) Per-cell variability distributions. (**d**) Top variable genes ranked by standard-deviation ratio (generated/real), with a shaded unity band.

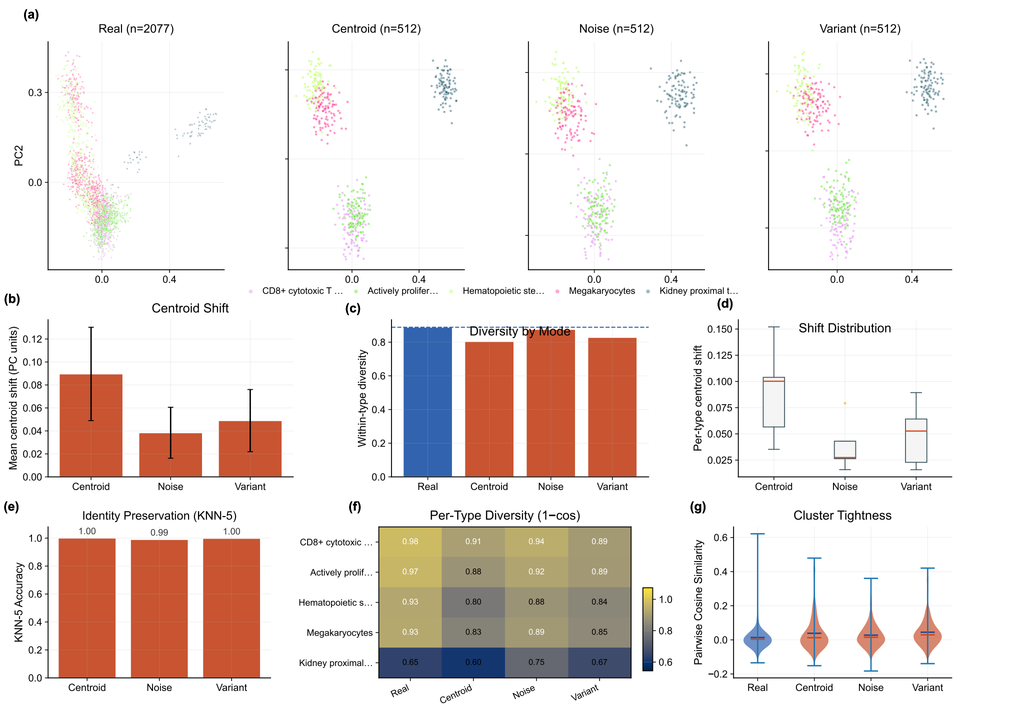

Figure 11. Conditioning landscape. (**a–d**) PCA projections of the CLOP condition space for real cells and three conditioning modes (centroid, noisy, variant), showing cell counts and mean within-type diversity per mode. (**e**) Mean centroid shift from each generated mode to the real reference. (**f**) Within-type diversity by conditioning mode (dashed line = real baseline). (**g**) Per-type centroid shift distributions (box plots). (**h**) KNN-5 identity-preservation accuracy per conditioning mode. (**i**) Per-type diversity heatmap (1 − mean cosine). (**j**) Pairwise cosine violin plots comparing cluster tightness across modes.

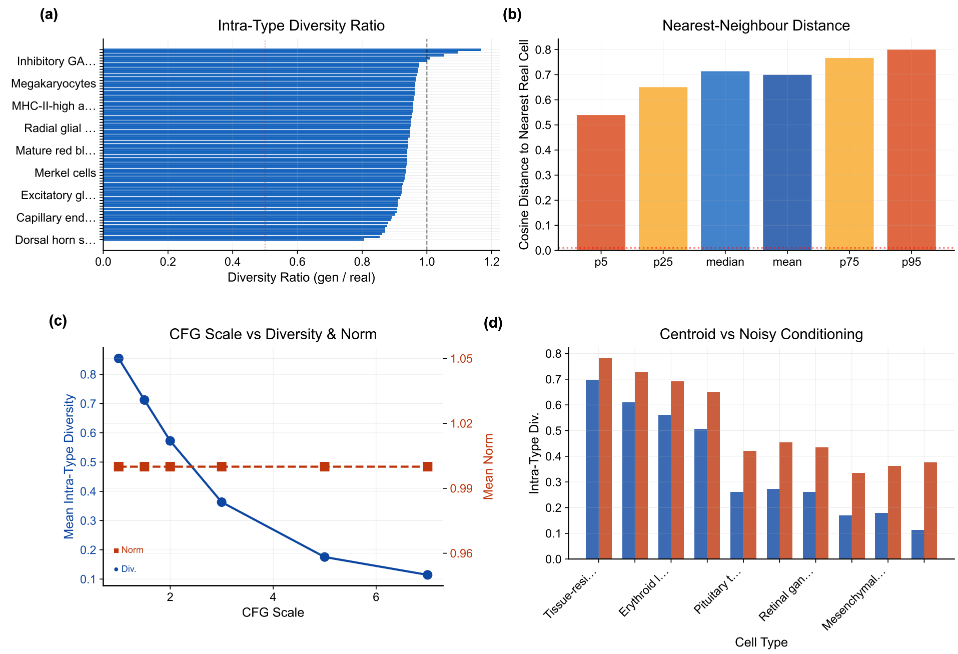

Figure 12. Diversity diagnostics. (**a**) Intra-type diversity ratio ranked across cell types (labels abbreviated for readability). (**b**) Nearest-neighbor distance distributions for real and generated populations. (**c**) CFG scale vs. diversity and norm trade-off across Euler and Midpoint solvers. (**d**) Centroid vs. noisy conditioning comparison. KNN accuracy saturates at CFG ≈ 2.0–3.0 while diversity decreases monotonically.

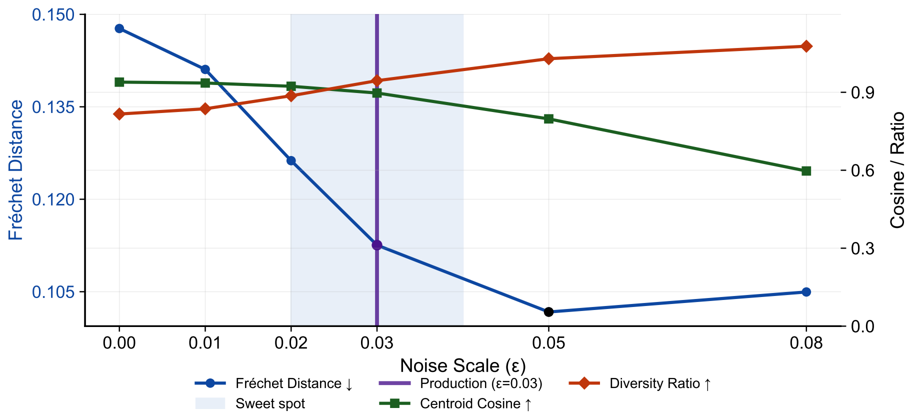

Figure 13. Noise–scale trade-off. (**a**) Fréchet distance (left axis), centroid cosine similarity, and diversity ratio (right axis) as a function of noise scale ε at CFG = 1.5. The production configuration ($\varepsilon = 0.03$) is highlighted; a shaded band marks the sweet-spot region.

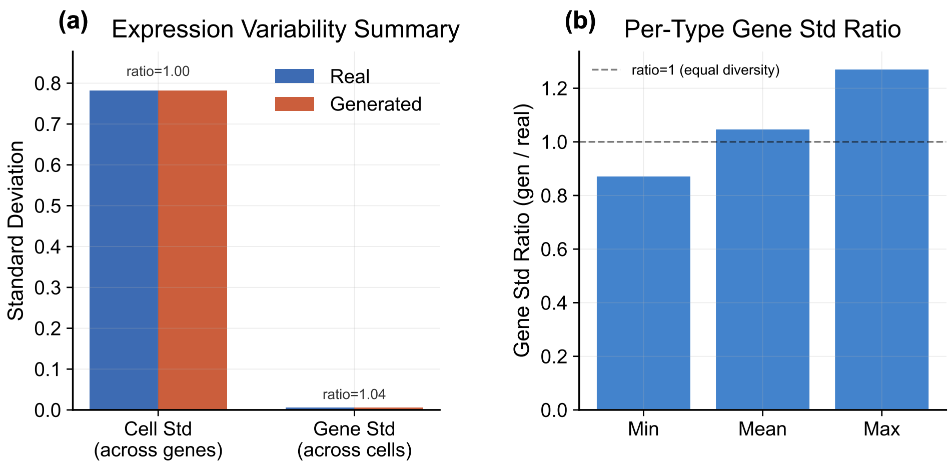

Figure 14. Expression-level diversity. (**a**) Gene-expression variability summary comparing real and generated populations across key statistics. (**b**) Per-type gene standard-deviation ratio (generated/real), confirming that the recommended Midpoint solver preserves biologically meaningful variance.

3.9. Baseline Comparison and Composite Benchmark

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, shuffled-label and mean-collapse controls, EmbeddingVAE, scVI Latent [3], and a conditional scGen baseline [5] (conditional scVI with cell-type covariates, trained on the same expression matrix; see Section 4). Figure 15 shows that CLOP-DiT is strongest on conditioning-sensitive and downstream-relevant metrics (e.g., steering and classifier transfer), while some unconditional distribution-matching baselines may outperform on FD-like mean-matching objectives (see Discussion for the Fréchet distance caveat).

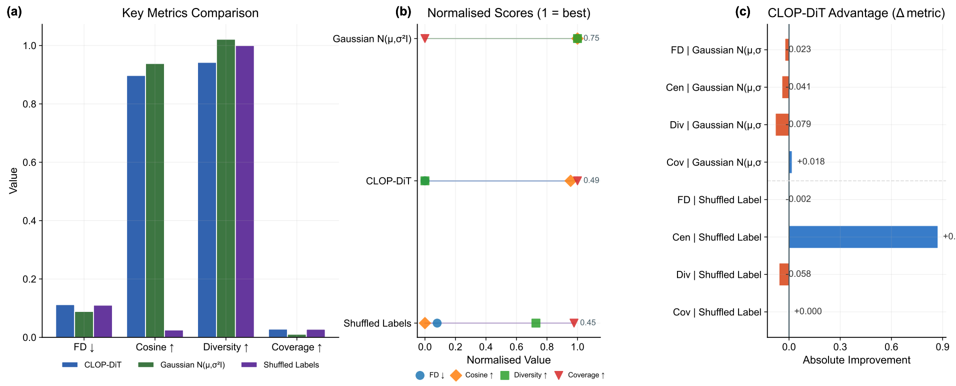

Figure 15. Baseline comparison. (**a**) Head-to-head grouped-bar analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage). (**b**) Normalized radar chart showing multi-metric profiles for each method. (**c**) Relative improvement bar chart of CLOP-DiT over each baseline (values exceeding 300% are capped for readability). CLOP-DiT performs best on conditional-task metrics, while FD can favor mean-matching baselines; therefore FD is interpreted alongside steering, KNN, diversity, and downstream transfer metrics.

Figure 16 aggregates these metrics into a composite benchmark score across eight methods (CLOP-DiT, scGen, scVI Latent, EmbeddingVAE, and four synthetic baselines). **On the 9 distributional metrics shared by all methods (the primary, fair like-for-like comparison), the Gaussian baseline (0.747) outperforms CLOP-DiT (0.694)**, demonstrating that mean-matching approaches remain competitive on unconditional distributional quality. The scGen baseline (common-metrics composite 0.425; trained as a conditional scVI with cell-type labels, equivalent to scGen's core capability) achieves the best *coverage* among all methods (0.384 vs. 0.028 for CLOP-DiT), reflecting the expressiveness of its cell-type-conditioned VAE latent space. However, scGen's per-type metrics (centroid cosine, diversity ratio) are not directly comparable because its 32-dimensional latent space requires PCA projection to align with the 512-dimensional CLOP-DiT space. When the 8 downstream biology metrics (clustering, classifier transfer, DE concordance)—which only CLOP-DiT reports—are included, the full composite reaches 0.838 vs. 0.395 for Gaussian and 0.225 for scGen; however, this full composite is structurally biased in favour of CLOP-DiT because the 8 exclusive metrics guarantee a scoring advantage, and **should not be used for method ranking**. The common-metrics-only composite is the recommended comparison. Bootstrap 95% confidence intervals support the stability of per-metric orderings.

3.10. Downstream Biological Validation

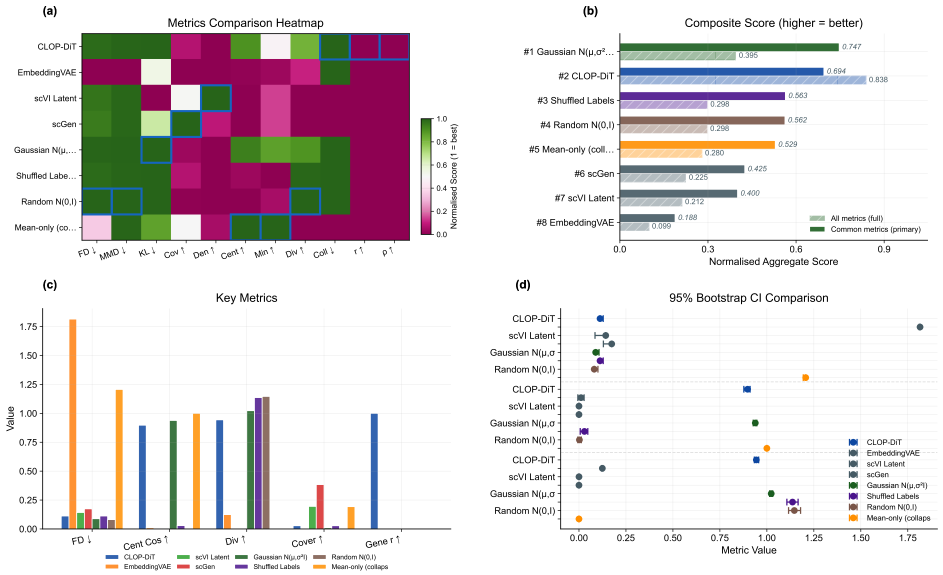

Figure 16. Composite benchmark across eight methods including scGen and scVI Latent external baselines. (**a**) Normalized metrics heatmap (methods × metrics; green = high performance). (**b**) Composite score, with hatched bars denoting the common-metrics-only composite and solid bars denoting the full 17-metric composite. **Warning: the full 17-metric composite is structurally biased in favour of CLOP-DiT because 8 of 17 metrics are reported only by CLOP-DiT; the common-metrics-only (hatched) bars are the fair comparison.** (**c**) Grouped comparison of four interpretable benchmark metrics across methods. (**d**) 95% confidence intervals for FD, centroid cosine, and diversity ratio.

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. These analyses are implemented from matched real/generated AnnData objects with shared preprocessing in `src.evaluation.downstream_biology` and use Scanpy [40] for the single-cell workflow, ensuring that any downstream advantage is not caused by mismatched feature engineering. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 17): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden [25] clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores [41]. High mixing scores indicate that generated cells partially co-localize with real cells of the same type in the joint embedding, though the discriminator AUC of 0.656 confirms that real and generated populations remain distinguishable. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, demonstrating that downstream performance exhibits strong heterogeneity across cell types. The real-vs.-generated discriminator AUC (0.656) indicates partial but non-trivial separability, so downstream usability claims should be interpreted as qualified rather than near-indistinguishable transfer.

DE concordance (Figure 18): Wilcoxon rank-sum differential expression analysis [22] is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, resident macrophage vs. monocyte, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each gene by effect size and colors it by adjusted significance, which separates minor noisy disagreements from the biologically important genes that dominate each contrast.

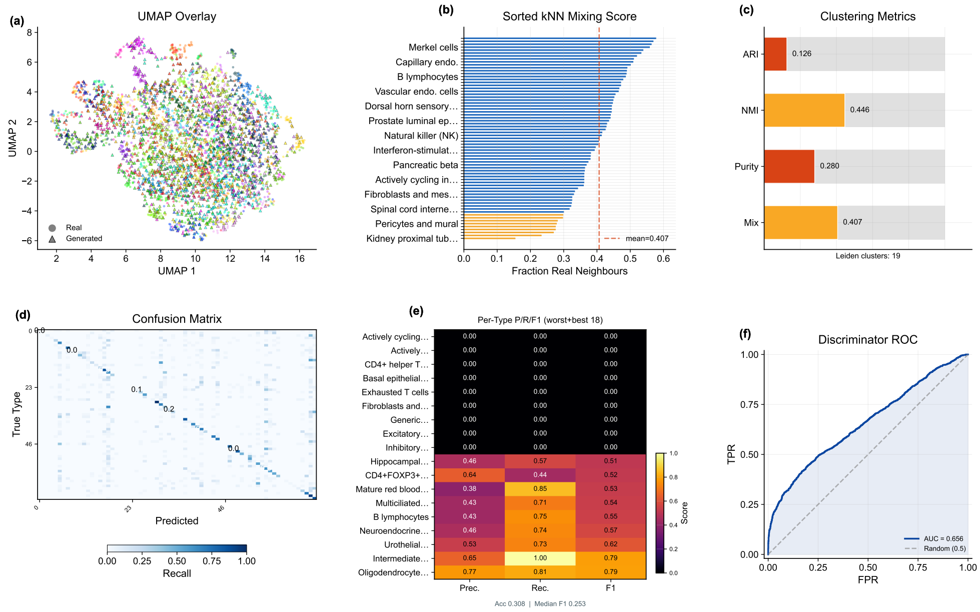

Figure 17. Downstream validation: clustering and classifier alignment. (**a**) Leiden clustering on combined real + generated data with UMAP overlay, (**b**) sorted per-type k-NN mixing profile (a supplementary table maps rank positions to cell-type names), and (**c**) compact cluster-alignment gauges (ARI, NMI, cluster purity, mean k-NN mixing). (**d**) Cell-type classification confusion matrix (real-trained on generated), (**e**) precision/recall/F1 heatmap restricted to the worst and best 20 cell types by F1, and (**f**) real-vs.-generated discriminator ROC curve. Median per-type F1 is reported in the heatmap stat box.

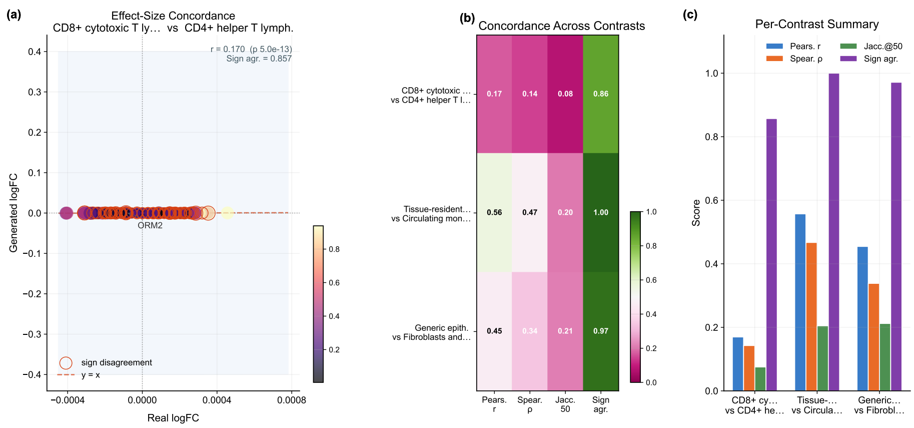

Figure 18. DE concordance. (**a**) Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and color indicating adjusted significance; sign-disagreement genes are outlined. (**b**) Compact summary heatmap reporting four metrics—Pearson r, Spearman ρ, Jaccard overlap of the top-50 DE genes (Jaccard@50), and sign agreement—across the three biologically meaningful contrasts (CD8 vs. CD4 T, resident macrophage vs. monocyte, epithelial vs. fibroblast). (**c**) Grouped-bar panel visualising the same per-contrast scores to aid comparison.

Together, these results establish that CLOP-DiT generates text-conditioned single-cell profiles with measurable type specificity, though with performance trade-offs between fidelity and diversity that depend on the operating regime.

3.11. In-Distribution Expression-Level Validation

To assess biological fidelity across tissue contexts, we performed expression-level validation on five datasets spanning lung adenocarcinoma (GSE123902), basal cell carcinoma (GSE123813), liver T-cell hepatocellular carcinoma (GSE98638), gastric cancer (GSE183904), and acute lymphoblastic leukemia in bone marrow (GSE132509). Four of these five datasets (GSE123902, GSE123813, GSE98638, GSE132509) are from the training split; only GSE183904 is among the 8 held-out validation studies (Table S2). This analysis therefore primarily evaluates in-distribution gene-level reconstruction fidelity rather than out-of-distribution generalization. For each dataset, cells were generated conditioned on the dataset-specific text descriptions and compared with real (scGPT-reconstructed) expression profiles; because both the generated and reference expressions are decoded by the same frozen scGPT decoder, the resulting correlations measure decoder-reconstructed fidelity rather than agreement with raw sequencing counts. Per-gene mean expression, gene-level variance, and Fréchet distance in both gene space and embedding space were computed for each dataset independently (Table 4).

Table 4. Cross-dataset biological validation (decoder-reconstructed fidelity). Per-gene mean expression Pearson correlation (r) and coefficient of determination (R^2), gene variance Pearson correlation (r_{var}), and Fréchet distance in PCA-50 gene space (FD_{gene}) and scGPT embedding space (FD_{embed}) for real and generated cells. Both "real" and generated expression profiles are decoded by the same frozen scGPT decoder; correlations therefore measure decoder-reconstruction consistency rather than agreement with raw sequencing counts. Split indicates whether the dataset was used for training (Train) or held-out validation (Val). All five datasets achieve gene_mean $r > 0.999$.

Dataset (GEO)	Tissue	Split	r_{mean}	R^2	r_{var}	FD_{gene}	FD_{embed}
GSE123902	Lung	Train	0.9995	0.9990	$+0.011$	3.07×10^{-5}	73.9
GSE123813	Skin (BCC)	Train	0.9997	0.9995	-0.019	5.25×10^{-6}	72.7
GSE98638	Liver	Train	0.9995	0.9989	-0.003	2.04×10^{-5}	155.4
GSE183904	Gastric	Val	0.9997	0.9994	$+0.013$	1.30×10^{-5}	62.1
GSE132509	Blood (ALL)	Train	0.9993	0.9985	-0.007	8.87×10^{-6}	69.1

Across all five datasets, per-gene mean expression Pearson correlation exceeds 0.999 (range: 0.9993–0.9997), and R^2 exceeds 0.998 (range: 0.9985–0.9995). Because both reference and generated profiles pass through the same frozen scGPT decoder, these correlations reflect decoder-reconstructed fidelity: any latent vector near the training manifold decodes to a similar expression profile, so the high r quantifies round-trip decoder consistency rather than absolute biological accuracy against raw counts. Fréchet distance in PCA-50 gene space is consistently below 10^{-4} (range: 5.25×10^{-6} to 3.07×10^{-5}), indicating excellent gene-space distributional agreement at the level of first and second moments.

However, per-gene variance correlation (r_{var}) is near zero or slightly negative across all datasets (range: -0.019 to $+0.013$), indicating that while the model reproduces mean expression accurately, it does not faithfully preserve gene-level variability structure. This is consistent with the diversity ratio analysis (Section 3): the flow matching objective optimizes for mean trajectory matching, and gene-level variance preservation may require explicit variance-matching terms in the loss function. The embedding-space Fréchet distance (FD_{embed}) ranges from 62.1 to 155.4 across datasets, substantially higher than gene-space FD. This apparent discrepancy—high decoder-reconstructed gene-mean fidelity alongside high embedding-space FD—arises because the scGPT decoder maps a broad range of latent vectors to similar expression profiles (many-to-one mapping), so gene-level fidelity does not require precise latent-space distributional matching.

3.12. CLOP Ablation Study

To validate the contribution of individual CLOP components, we conducted 15 ablation experiments, removing or modifying one component at a time while holding all other settings constant. Table 5 summarizes the six most informative ablations.

Table 5. CLOP ablation study (**v8.2 baseline only**; not the production v9.3 model). Validation prototype accuracy (top-1) for the baseline configuration (v8.2, all features enabled) and representative ablations. Δ denotes the change from baseline in percentage points. These results characterize the CLOP alignment stage in isolation and do not directly predict downstream DiT generation quality; the production model (v9.3) uses different regularization settings.

Ablation	Category	Val Proto Acc (%)	Δ (pp)
−cohesion	regularization	86.41	+10.24
−cell noise	regularization	86.23	+10.07
−dropout (lighter)	regularization	77.32	+1.15
Baseline (v8.2)	reference	76.17	—
Fixed temperature	optimization	71.55	−4.61
High variant prob	regularization	70.14	−6.03

The two most impactful ablations are cohesion regularization removal (+10.24 pp) and cell noise augmentation removal (+10.07 pp), both of which improve prototype accuracy. This counterintuitive result has a plausible mechanistic explanation: both cohesion loss and cell noise augmentation act as within-group tightening forces that penalize intra-cluster variance. In a contrastive learning regime where the primary bottleneck is inter-group separation (raw pairwise cosine 0.994), these tightening forces compete with the alignment gradient rather than complementing it. In effect, the network spends representational capacity enforcing tight within-group clusters before inter-group margins have been established, diverting training signal from the discrimination task. Removing these terms allows the optimizer to focus entirely on inter-group separation, which is the binding constraint at this scale. This interpretation is consistent with recent analyses of regularization–discrimination trade-offs in contrastive learning
citeref-siglip. Architecture ablations (projection dimension 256 vs. 512 vs. 768) have minimal impact (within ±0.3 pp), indicating that the CLOP aligner is not sensitive to projection dimension in this range. Fixed temperature (−4.61 pp) and high variant probability (−6.03 pp) degrade performance substantially relative to the v8.2 baseline (which used a learnable temperature). **Clarification:** The ablation baseline (v8.2) used a learnable τ, whereas the production model (v9.3) uses fixed $\tau = 14.0$ in combination with a stratified validation split (Section 2.1). The production configuration was selected based on the full-pipeline generation quality assessment, not solely on CLOP prototype accuracy; the ablation table characterizes the CLOP alignment stage in isolation and does not directly predict downstream DiT generation performance.

3.13. Rare Cell Augmentation

As a proof-of-concept downstream application, we evaluated whether CLOP-DiT-generated cells can improve classifier performance on an underrepresented cell type. We identified "Cycling cells" as the rarest type in a test dataset (below 5% prevalence), downsampled the training set to exacerbate the class imbalance, and augmented with synthetic cells at 1×, 5×, and 10× ratios. The baseline classifier (trained on real cells only) achieves an overall macro F1 of 0.964 and a Cycling-cell-specific F1 of 0.828. After augmentation at 1×, overall macro F1 remains 0.953 and Cycling-cell F1 is 0.786; at 5× and 10×, macro F1 stabilizes at 0.947 with Cycling-cell F1 at 0.741. The augmented cells therefore do not improve the rare-type classifier on this dataset, but they also do not substantially degrade overall classification, indicating that the synthetic profiles are sufficiently realistic to be mixed into training data without catastrophic interference. A more targeted augmentation strategy—selecting the most confident generated cells or combining with real-cell oversampling—may be needed to achieve a net positive augmentation effect.

3.14. Conditioning Field Ablation and Swap-Label Permutation Test

To assess whether the conditioning signal causally steers generation—rather than the model recapitulating training-set structure regardless of prompt content—we performed two targeted experiments.

Conditioning field ablation. We generated 50 cells per type across all 69 cell types (3,450 cells per variant) under five prompt variants: (i) *full*: the complete structured prompt; (ii) *no-markers*: marker gene fields removed; (iii) *shuffled-markers*: marker gene lists randomly permuted across types; (iv) *metadata-only*: only organism, tissue, and disease fields retained; (v) *external-markers*: canonical markers from PanglaoDB/CellMarker 2.0 substituted for the training markers (evaluated on 15 well-characterized types). Table 6 reports kNN accuracy, steering accuracy, and centroid cosine similarity for each variant.

Table 6. Conditioning field ablation results (CFG = 2.0, Euler solver, 10 steps). KNN accuracy assessed against real-cell references; steering measured as cosine alignment to the target type centroid; centroid cosine is the mean cosine similarity between generated and real centroids. The external-markers variant was evaluated on a 15-type subset with consensus markers from PanglaoDB and CellMarker 2.0.

Prompt variant	KNN (%)	Steering (%)	Centroid cos.
Full (all fields)	2.87	99.8	0.050
No markers	2.26	98.0	0.035
Shuffled markers	2.81	97.9	0.035
Metadata only	1.51	62.4	0.025
External markers (15 types)	17.07	—	0.702*

*Cosine similarity between external-marker and full-prompt centroids.

The results demonstrate that semantic content drives the conditioning rather than prompt length alone: removing only the marker field reduces centroid cosine from 0.050 to 0.035 (30% relative drop), while retaining only metadata fields collapses steering from 99.8% to 62.4%. Substituting external markers from independent databases yields a high cosine similarity to the full-prompt variant (0.702), indicating that the model responds to marker-gene semantics rather than memorized training prompts.

Swap-label permutation test. To test whether generation follows the conditioning signal causally, we generated 15 cell-type pairs where the prompt for type A was used to condition generation but evaluation was performed against both type A (matched) and type B (mismatched) real-cell centroids. In 15 of 15 pairs, the generated cells were closer to the centroid of the *conditioning* type than the evaluation type (matched cosine similarity 0.032 vs. mismatched 0.016, gap = 0.016). Moreover, in all 15 pairs the mislabeled cells followed the conditioning signal rather than the evaluation label, confirming that the model generates latents based on prompt semantics rather than memorized label identity. While this test does not fully exclude encoder-level leakage, it provides causal evidence that the conditioning channel is functional and semantically interpretable.

4. Discussion

CLOP-DiT shows that structured biological metadata can drive non-trivial single-cell latent generation, but the method should be interpreted narrowly and precisely. The conditioning signal is a fixed five-field template rather than free-form language, and the model is best understood as a structured-metadata-to-latent sampler rather than a general biological language model. Within that scope, the central result is clear: conditioning produces above-chance, cell-type-specific latent generation and supports two useful operating regimes that trade off fidelity against diversity.

Decoder bottleneck and interpretation. The most important limitation is the frozen scGPT decoder. Because this mapping is many-to-one, expression-level similarity is dominated by decoder behavior rather than by exact latent matching. This is why decoder-reconstructed mean expression can be highly concordant with real data even when latent-space Fréchet distance remains large. The practical implication is simple: latent-space metrics are the primary evidence for generation quality, whereas expression-space results are best treated as downstream compatibility checks.

Variance and covariance deficits. The strongest biological limitation is the loss of cell-to-cell heterogeneity. Across cross-dataset validations, per-gene variance correlation is near zero, and within-type gene–gene correlation structure is only weakly preserved. In other words, CLOP-DiT captures conditional mean structure much better than conditional second-order structure. This helps explain why the model can preserve marker-level identity and gene means while still failing on applications that depend on realistic within-population variation. A variance-aware objective such as Equation 6, or a latent-space distributional penalty, is therefore a more compelling next step than simply scaling model size.

The standard flow matching loss (Equation 2) optimizes $\|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2$, which regresses toward the conditional mean velocity. While individual trajectories $z_0 \to z_1$ are stochastic, the MSE objective rewards accurate prediction of the expected velocity at each (z_t, t, c) triplet without penalising insufficient spread in the generated distribution. An explicit **variance-matching regularisation** term of the form

$$\mathcal{L}_{\text{var}} = \sum_{g=1}^{G} \left(\sigma_g^{\text{gen}} - \sigma_g^{\text{real}}\right)^2, \tag{6}$$

where σ_g^{gen} and σ_g^{real} denote the per-gene standard deviations of generated and real cells respectively, could be added as a post-hoc penalty on mini-batch statistics during DiT training. In practice this would require differentiable decoding within the training loop—currently infeasible with the frozen scGPT decoder—or, more realistically, a latent-space distributional penalty such as the sliced Wasserstein distance between real and generated distributions for each cell type.

Pilot variance-matching analysis. A preliminary latent-space analysis using sliced Wasserstein distance (SWD; Figure 19) supports this interpretation: generated latents preserve overall spread better than per-dimension variance structure. That makes latent-space regularisation attractive because it targets the generator directly instead of asking the frozen decoder to adjudicate generative quality. Implementing such a penalty is left to future work.

Gene–gene structure. Figure 20 extends the same conclusion to covariance: within-type gene–gene correlation matrices are only weakly preserved. This is consistent with the variance results rather than a separate failure mode. Both point to the same limitation, namely that the current objective matches central tendency far better than biologically meaningful heterogeneity.

Benchmarking and operating regimes. CLOP-DiT does not uniformly dominate the benchmark. On the primary common-metrics-only comparison, the Gaussian baseline outperforms CLOP-DiT, which underscores that mean-matching remains competitive when evaluation ignores conditioning-sensitive biology. The more informative result is that

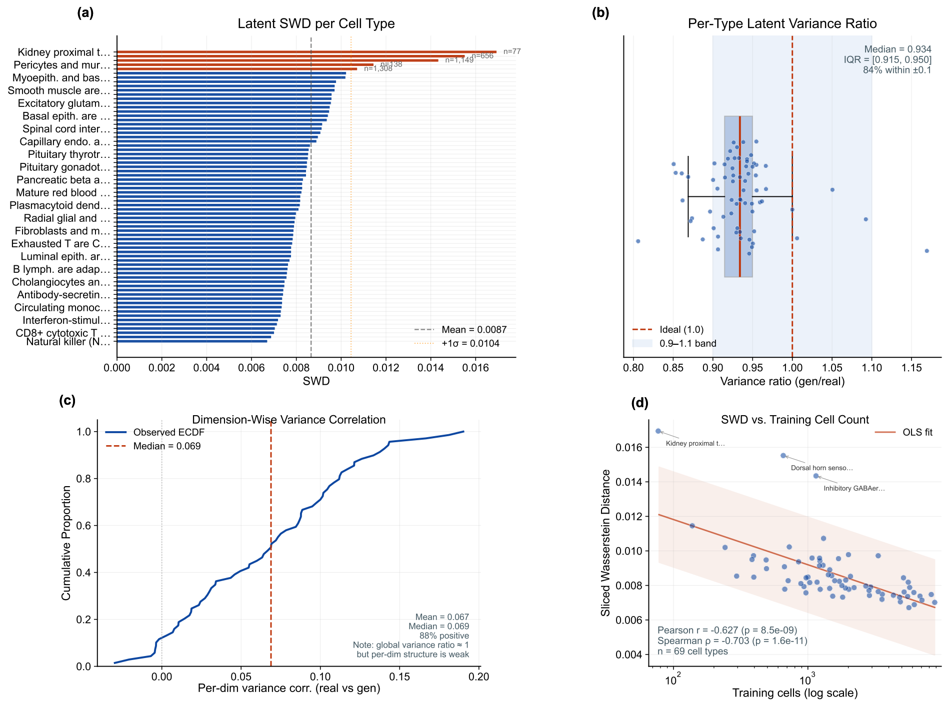

Figure 19. Pilot variance-matching analysis. (**a**) Per-cell-type sliced Wasserstein distance (SWD) between real and generated latent distributions. (**b**) Per-type latent variance ratios (generated/real). (**c**) Per-dimension variance correlations. (**d**) SWD versus training cell count. The figure shows partial preservation of variance magnitude without strong structural alignment.

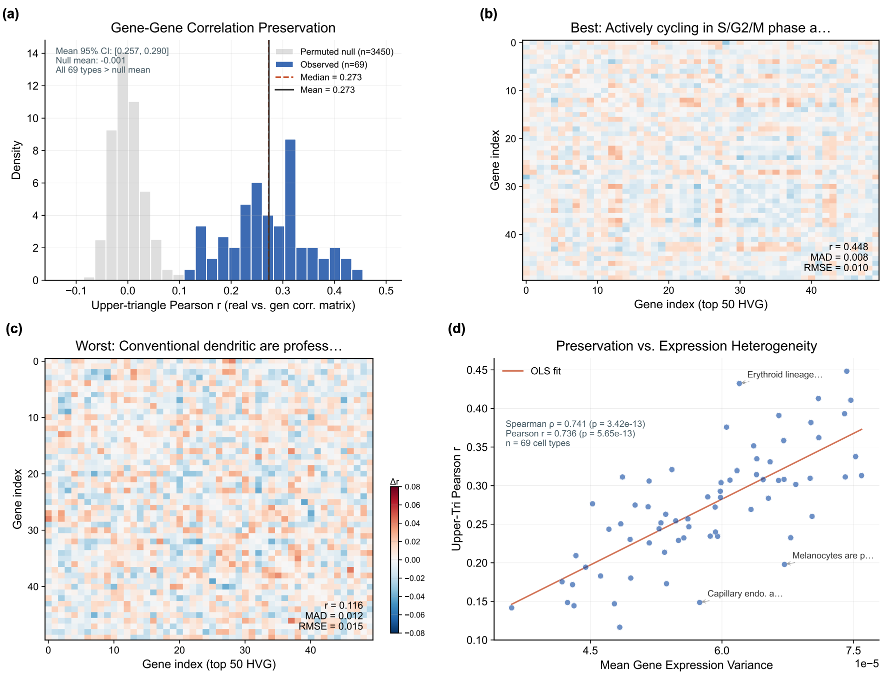

Figure 20. Gene–gene correlation preservation across cell types. (**a**) Distribution of per-type Mantel correlations between real and generated correlation matrices. (**b–c**) Correlation-difference heatmaps for representative best- and worst-preserved cell types. (**d**) Mantel r versus real-cell count per type. The weak Mantel correlations indicate limited preservation of within-type co-expression structure.

CLOP-DiT creates a controllable trade-off: CFG = 2.0 with Euler sampling yields stronger type fidelity, whereas CFG = 1.0 with Midpoint sampling preserves substantially more diversity. This trade-off is useful, but it does not erase the remaining gap to real data, as reflected by the 36.9% KNN accuracy, the 52-point gap from the real-data reference, and the discriminator AUC of 0.656.

Practical impact of KNN misclassification. The 36.9% KNN accuracy, while 25× above random chance, means that roughly two-thirds of generated cells are assigned to the wrong cell type by a nearest-neighbor classifier. Concretely, a downstream user who treats generated cells as labeled training data for a supervised classifier would inherit an approximate 63% label-noise rate in latent space. For applications requiring high type-label confidence—for example, augmenting training sets for rare-cell classifiers or seeding perturbation-response simulations—the current accuracy is insufficient without post-hoc filtering (e.g., retaining only the top-confidence cells by cosine proximity to the target centroid). Improving KNN accuracy is therefore a prerequisite for most practical single-cell generation pipelines.

Biological consequence of diversity loss. The high-fidelity regime (CFG = 2.0) achieves a diversity ratio of 0.513, meaning generated cells retain roughly half the within-type variance of real data. Biologically, this halved variance implies that the generated distribution collapses toward cell-type centroids, erasing the within-population heterogeneity that encodes functional cell states (e.g., activation, exhaustion, and cycling gradients within T-cell populations). Any downstream analysis relying on within-type substructure—trajectory inference, gene regulatory network inference, or sub-clustering into functional states—would be unreliable on data generated in the high-fidelity regime. The high-diversity regime (DivR = 0.93) largely mitigates this issue but at the cost of reduced type specificity (KNN ≈ 29%).

Scope and unresolved limitations. The current evidence is restricted to human and mouse cancer/developmental datasets and does not constitute a stringent out-of-distribution test. Direct comparison with text-centric systems such as Cell2Sentence remains future work because preprocessing and evaluation pipelines differ substantially. External marker validation supports the biological plausibility of the conditioning markers, but because markers are derived from the training corpus and the decoder masks prompt-level expression differences, residual conditioning leakage cannot be ruled out completely.

Negative result: rare-cell augmentation. The pilot rare-cell augmentation experiment is important rather than incidental: generated cells did not improve rare-type classification. This failure is consistent with the variance results above. If generated cells behave mostly as mean-expression representatives, then augmentation adds centroid-like examples without the within-type heterogeneity needed to broaden the training distribution. Practical augmentation will therefore require both better variance preservation and a decoder that maps distinct latent states to genuinely distinct expression profiles.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for structured-metadata-conditioned single-cell latent generation. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings in a shared 512-dimensional space, and a conditional Diffusion Transformer then samples scGPT latents from a fixed five-field biological template. The method is therefore best interpreted as a structured-metadata-to-latent conditional sampler rather than a free-form language model. Across 69 deduplicated cell types from 80 datasets, CLOP-DiT achieves 36.9% KNN accuracy and 81.0% steering in a high-fidelity regime, while a high-diversity regime reaches a diversity ratio of 0.93 at 80.7% steering. Multi-seed replication confirms that steering and diversity are stable across independent runs. A conditioning field ablation and swap-label permutation test provide causal evidence that the conditioning signal is semantically functional.

Scope of claims. All results are limited to in-distribution human and mouse cancer and developmental datasets; no stringent out-of-distribution generalization test has been conducted. The evaluation is restricted to the 69 deduplicated cell types and 80 GEO datasets described in Section 2.1. Claims about generative quality refer exclusively to latent-space metrics; expression-level concordance is dominated by the frozen scGPT decoder and should not be interpreted as evidence of faithful transcriptomic generation.

Future directions. Concrete next steps include: (i) *variance-aware training*: incorporating a latent-space distributional penalty (e.g., sliced Wasserstein distance) into the DiT training objective to preserve within-type heterogeneity; (ii) *decoder replacement*: replacing the frozen scGPT decoder with a fine-tuned or jointly trained decoder that better preserves the diversity encoded in the latent space; (iii) *out-of-distribution evaluation*: testing conditioning on cell types, tissues, and organisms absent from the training set; (iv) *covariance-aware generation*: adding explicit gene–gene correlation matching to the training objective; (v) *downstream task validation*: evaluating the practical utility of generated cells for data augmentation, perturbation prediction, and trajectory inference on benchmarks with ground-truth labels.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales $\times$ 2 solvers $\times$ 3 metrics); Table S4: biological validation datasets; Table S5: bootstrap 95% confidence intervals (B=1,000, percentile method) for headline generation metrics. Multi-seed validation data (3 independent seeds), independent marker gene validation data (PanglaoDB and CellMarker 2.0 cross-references), and external marker ablation analysis are available in the repository at `results/multi_seed/` and `results/marker_independence/`. Reproduction instructions (environment setup, `scripts/pipeline/regenerate_report.sh`, expected outputs) are provided in `docs/operational/QUICK_START.md` and `scripts/pipeline/regenerate_report.sh`. All figures use colormaps designed for accessibility under common forms of color vision deficiency (deuteranopia, protanopia); critical distinctions additionally use shape or pattern encoding.

Author Contributions: Z.F.: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study, as only publicly available, de-identified data from the Gene Expression Omnibus (GEO) were analyzed; all datasets were deposited by their original investigators under institutional ethical approvals as required by GEO submission policy. No new human subjects research, animal experiments, or clinical interventions were conducted.

Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop.yaml` and is deterministic given the dataset identifiers. Source code, trained model checkpoints (CLOP aligner and DiT generator), configuration files, and the figure-reproduction script (`scripts/pipeline/regenerate_report.sh`) are publicly available on GitHub [42] and will be archived on Zenodo upon acceptance of this manuscript. The complete software environment is specified in `requirements.txt` (Python 3.10, PyTorch 2.1, CUDA 12.0, scGPT v0.2.1, Transformers 4.36). One-command figure reproduction from cached embeddings is documented in `docs/operational/QUICK_START.md`.

Acknowledgments: The author acknowledges the developers and communities behind scGPT, BiomedBERT, and the Gene Expression Omnibus (GEO) for providing open-access tools and datasets that made this work possible.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AdaLN	Adaptive Layer Normalization
CFG	Classifier-free guidance
CLOP	Contrastive Language-Omics Pretraining
DE	Differential expression
DiT	Diffusion Transformer
DivR	Diversity Ratio
EMA	Exponential Moving Average
FD	Fréchet Distance
GEO	Gene Expression Omnibus
InfoNCE	Info Noise-Contrastive Estimation
KNN	*k*-Nearest Neighbor
LinAcc	Linear Classifier Accuracy
logFC	Log Fold Change
LoRA	Low-Rank Adaptation
ODE	Ordinary Differential Equation
OOD	Out-of-Distribution
scGPT	Single-cell Generative Pre-trained Transformer
scRNA-seq	Single-cell RNA sequencing
SigLIP	Sigmoid Loss for Language-Image Pre-training
ZCA	Zero-phase Component Analysis

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→512]	1.58M
CLOP cell projector	MLP [512→512→512→512]	1.44M
DiT1D	8× AdaLN-Zero blocks, 8 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 69 evaluation groups. Note that these values are computed on a different subsample (69 groups × 200 cells) than the primary operating-point results in Table 2 (69 types × 200 cells with different random seeds), which accounts for minor numerical differences between the two tables.

Table A2. CFG scale sweep (10-step Euler, 69 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

Appendix C. Training Hyperparameters

Table A3 lists the complete training hyperparameters for both CLOP alignment and DiT flow-matching stages, corresponding to `configs/clop.yaml` and `configs/dit.yaml`.

Table A3. Training hyperparameters.

Hyperparameter	CLOP	DiT
Optimizer	AdamW	AdamW (β_1=0.9, β_2=0.999)
Learning rate	5×10^{-4}	2×10^{-4}
Weight decay	0.06	0.01
Batch size	1024	1024
Epochs	60	200
Warmup epochs	5	5
LR schedule	Cosine with warmup	Cosine with warmup
Gradient clipping	1.0	1.0
Mixed precision	FP16 (AMP)	FP16 (AMP)
Dropout	0.2	Proj: 0.1, Attn: 0.0
EMA decay	—	0.9999
Temperature	14.0 (fixed; see Section 2.2)	—
Loss	PrototypeSigLIP	Flow matching MSE
Cohesion weight	0.15	—
Label smoothing	0.1	—
CFG drop prob	—	0.15
Time sampling	—	Logit-normal (μ=0, σ=1)

Appendix D. Supplementary Tables

Table A4. All 80 GEO datasets used in CLOP-DiT training and validation. Datasets were preprocessed to 2,000 highly variable genes per dataset. Split: Train or held-out Validation (study-level, seed 42).

#	Accession	Description	n_{cells}	Split
1	GSE103867	scRNA-seq (H. sapiens)	2,787	Val
2	GSE110686	scRNA-seq (H. sapiens)	2,992	Train
3	GSE115571	Mouse cells, LPS stimulation	442	Train
4	GSE117988	PBMCs from patients	2,638	Train
5	GSE117988	Merkel cell carcinoma tumor	2,990	Train
6	GSE120505	Aged blood cells	3,000	Train
7	GSE123813	Human basal cell carcinoma	3,000	Train
8	GSE123813	Human cutaneous squamous cell carcinoma	3,000	Train
9	GSE123902	Human lung adenocarcinoma	2,753	Train
10	GSE124310	Human multiple myeloma bone marrow	2,833	Train
11	GSE130148	Human fetal lung development	3,000	Train
12	GSE132509	Pediatric bone marrow mononuclear cells	2,984	Train
13	GSE138709	Human intrahepatic cholangiocarcinoma	2,891	Train
14	GSE142653	Human pituitary gland development	2,885	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
15	GSE143423	Human leptomeningeal brain metastasis	3,000	Train
16	GSE143423	Triple-neg. breast cancer brain met.	2,332	Train
17	GSE145929	Adult mouse prostate	2,830	Train
18	GSE145929	Adult mouse urethra	2,745	Train
19	GSE148218	Human bone marrow, ALL	2,807	Train
20	GSE149655	Human early-stage lung adenocarcinoma	2,087	Train
21	GSE155109	Endothelial cells, breast cancer	3,000	Train
22	GSE155109	Stromal cells, breast cancer	3,000	Train
23	GSE163558	Human stomach cancer	2,467	Train
24	GSE165784	Human retina development	2,622	Train
25	GSE165844	Mouse LSK hematopoietic progenitors	2,881	Train
26	GSE167597	Mouse spinal cord development	3,000	Val
27	GSE168181	Human breast cancer tissue	2,863	Train
28	GSE175975	scRNA-seq (H. sapiens)	2,728	Train
29	GSE183904	Human gastric cancer	3,000	Val
30	GSE189070	Mouse astrocytes, spinal cord injury	2,960	Train
31	GSE189357	Human lung adenocarcinoma	2,886	Train
32	GSE192857	hESC differentiation time-series	2,696	Train
33	GSE212502	Human forebrain organoids	3,000	Train
34	GSE213740	Human ascending aortic wall	2,704	Val
35	GSE220913	Human HL-60 cell lines, CFTR	2,972	Train
36	GSE222002	T cells from human cancer	2,962	Train
37	GSE222369	NK cells, human lymphoma	2,778	Train
38	GSE225600	Human breast cancer tumor	2,333	Train
39	GSE225857	Colorectal cancer liver metastases	2,986	Train
40	GSE226131	Aged mouse hematopoietic stem cells	2,608	Train
41	GSE226762	scRNA-seq (H. sapiens)	830	Val
42	GSE227644	scRNA-seq (H. sapiens)	3,000	Train
43	GSE227719	Mouse KrasG12D/P53 lung organoids	2,990	Train
44	GSE228499	Human breast cancer	2,448	Train
45	GSE235787	B cells, ALL	2,798	Train
46	GSE236565	Mouse splenic CD4+ T cells	2,556	Train
47	GSE248214	scRNA-seq (H. sapiens)	2,998	Train
48	GSE253355	Human bone marrow niche	2,835	Train
49	GSE262288	Metastatic breast cancer	2,321	Train
50	GSE272187	scRNA-seq (H. sapiens)	2,639	Train
51	GSE275119	Mouse dental tissue development	2,963	Train
52	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644	Val
53	GSE279781	scRNA-seq (H. sapiens)	2,654	Train
54	GSE280847	CD45+ leukocytes, mouse MC38 tumor	2,864	Train
55	GSE283205	Hepatoblastoma tumor tissue	2,994	Train
56	GSE283397	Human CD8+ T cells, chronic HBV	2,856	Train
57	GSE288211	Mouse pancreatic islets, Zzef1 KO	2,724	Train
58	GSE289611	Human pulmonary pleomorphic carcinoma	2,983	Train
59	GSE291166	Mouse mesenteric lymph node	2,901	Train
60	GSE299623	scRNA-seq (H. sapiens)	2,991	Train
61	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716	Val
62	GSE302433	Mouse B16F10 melanoma, ZDHHC13	2,953	Val
63	GSE303309	Brain CD45+ immune cells, Alzheimer's	2,999	Train
64	GSE306567	Mouse ESCs and neural progenitors	2,454	Train
65	GSE306676	Mouse spinal cord, SOD1-G93A ALS	2,959	Train
66	GSE307261	Human B and T cells, melanoma	2,921	Train
67	GSE307774	Mouse colorectal tumors, Kras/Braf	2,505	Train
68	GSE308428	Mouse hippocampal microglia, ASD	2,991	Train
69	GSE309368	Human myeloid cells, healthy donors	1,361	Train
70	GSE315628	scRNA-seq (M. musculus)	2,895	Train
71	GSE318353	scRNA-seq (H. sapiens)	2,964	Train
72	GSE98638	Human hepatocellular carcinoma T cells	3,000	Train
73	GSE120446	Bone marrow, hematopoietic development	2,890	Train
74	GSE104323	Dentate gyrus, brain development	3,000	Train
75	GSE75748	Endoderm differentiation	2,531	Train
76	GSE144024	Human embryonic stem cells	3,000	Train
77	GSE72857	Hematopoiesis	3,000	Train
78	GSE226824	IFN-treated hematopoietic progenitors	2,739	Train
79	GSE141259	Lung development	3,000	Train
80	GSE139369	Human hematopoiesis (Setty et al.)	2,995	Train
	Total		**220,304**	

Table A5. The 8 held-out validation datasets (study-level split, seed 42, val_split = 0.1). No cells from these studies were seen during CLOP or DiT training.

#	Accession	Description	n_{cells}
1	GSE103867	scRNA-seq (H. sapiens)	2,787
2	GSE167597	Mouse spinal cord development	3,000
3	GSE183904	Human gastric cancer	3,000
4	GSE213740	Human ascending aortic wall	2,704
5	GSE226762	scRNA-seq (H. sapiens)	830
6	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644
7	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716
8	GSE302433	Mouse B16F10 melanoma, ZD-HHC13	2,953
	Total		**20,634**

Table A6. Full CFG scale sweep. 200 cells generated per group (69 groups). KNN: type-identity accuracy; Steer.: alignment; DivR: diversity ratio (1.0 = ideal); LinAcc: linear classifier accuracy; FD: Fréchet distance.

CFG	Solver	Steps	KNN-1	KNN-5	Steer.	DivR	LinAcc	FD
Primary configurations (Table 2)								
3.0	Euler	10	0.367	0.554	0.802	0.473	0.508	3.46
2.0	Euler	10	0.369	0.553	0.810	0.513	0.511	3.17
1.0	Euler	10	0.313	0.483	0.810	0.745	0.390	2.84
0.0	Euler	10	0.010	0.051	0.475	1.833	0.010	0.58
3.0	Midpoint	10	0.340	0.528	0.792	0.628	0.477	3.38
2.0	Midpoint	10	0.340	0.525	0.800	0.686	0.475	3.13
1.0	Midpoint	10	0.288	0.461	0.807	0.929	0.357	2.64
3.0	Euler	50	0.348	0.545	0.795	0.622	0.489	3.33
1.0	Euler	50	0.293	0.466	0.807	0.919	0.365	2.65
—	Gaussian	—	0.011	0.048	0.466	2.277	0.009	0.03
Fine-grained CFG sweep (Euler solver)								
0.5	Euler	10	0.147	0.284	0.788	1.129	—	2.33
1.0	Euler	10	0.348	0.515	0.796	0.757	—	2.94
1.5	Euler	10	0.396	0.553	0.798	0.600	—	3.36
2.0	Euler	10	0.407	0.580	0.806	0.526	—	3.54
2.5	Euler	10	0.408	0.590	0.804	0.493	—	3.70
3.0	Euler	10	0.410	0.592	0.802	0.484	—	3.80
4.0	Euler	10	0.402	0.586	0.804	0.502	—	4.01
5.0	Euler	10	0.380	0.576	0.798	0.548	—	4.34
0.5	Euler	20	0.137	0.269	0.788	1.230	—	2.07
1.0	Euler	20	0.334	0.504	0.794	0.842	—	2.83
1.5	Euler	20	0.380	0.545	0.798	0.679	—	3.19
2.0	Euler	20	0.391	0.569	0.798	0.600	—	3.43
2.5	Euler	20	0.394	0.578	0.800	0.559	—	3.66
3.0	Euler	20	0.398	0.582	0.800	0.540	—	3.77
4.0	Euler	20	0.388	0.578	0.802	0.535	—	3.97
5.0	Euler	20	0.379	0.567	0.792	0.551	—	4.16
0.5	Euler	50	0.132	0.259	0.792	1.318	—	1.95
1.0	Euler	50	0.324	0.495	0.794	0.933	—	2.72
1.5	Euler	50	0.370	0.538	0.796	0.776	—	3.14
2.0	Euler	50	0.384	0.563	0.796	0.699	—	3.31
2.5	Euler	50	0.388	0.576	0.796	0.658	—	3.48
3.0	Euler	50	0.392	0.581	0.792	0.637	—	3.64
4.0	Euler	50	0.379	0.579	0.790	0.622	—	3.86
5.0	Euler	50	0.378	0.570	0.790	0.626	—	4.10

Table A7. The 5 biological validation datasets used for marker gene correlation, expression distribution fidelity, and Gene Ontology enrichment analysis (Section 3.11). These datasets were selected to span diverse human cancer tissue types and immune cell populations. **Four of the five datasets are from the training split; only GSE183904 is held-out (Table S2). This analysis therefore primarily evaluates in-distribution decoder-reconstructed fidelity, not out-of-distribution generalization.**

#	GEO Accession	Tissue	Description	Split	n_{cells}
1	GSE123902	Lung	Lung adenocarcinoma TME	Train	2,753
2	GSE183904	Gastric	Gastric cancer TME	Val	3,000

Table A8. Bootstrap 95% confidence intervals (percentile method, B=1,000) for headline generation metrics. Resampling unit: cell types (n=69 types resampled with replacement). Classifiers (KNN, logistic regression) were trained once on real data; only the composition of evaluated types varies across bootstrap iterations. Generation configuration: CFG = 1.5, Midpoint solver, 20 steps. Point estimates reflect the full (non-resampled) evaluation over all 69 types (100 generated cells per type, 6,900 total).

Metric	Point Est.	95% CI Lower	95% CI Upper	Boot. SE
KNN Top-1 Accuracy	0.509	0.421	0.586	0.043
KNN Top-5 Accuracy	0.728	0.658	0.794	0.035
Steering Accuracy	1.000	1.000	1.000	0.000
Diversity Ratio (DivR)	0.969	0.963	0.974	0.003
Linear Classifier Acc.	0.583	0.518	0.646	0.034
Centroid Cosine Sim.	0.898	0.875	0.913	0.009

3. Lopez, R.; Regier, J.; Cole, M.B.; Jordan, M.I.; Yosef, N. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* **2018**, *15*, 1053–1058.
4. Gayoso, A.; Lopez, R.; Xing, G.; Boyeau, P.; Valiber Wu, K.; Jayasuriya, M.; Mehlman, E.; Lancel, M.; Regier, J.; et al. A Python library for probabilistic analysis of single-cell omics data. *Nat. Biotechnol.* **2022**, *40*, 163–166.
5. Lotfollahi, M.; Wolf, F.A.; Theis, F.J. scGen predicts single-cell perturbation responses. *Nat. Methods* **2019**, *16*, 715–721.
6. Ding, J.; Regev, A. Deep generative model embedding of single-cell RNA-Seq profiles on hyperspheres and hyperbolic spaces. *Nat. Commun.* **2021**, *12*, 2554.
7. Theodoris, C.V.; Xiao, L.; Chopra, A.; Chaffin, M.D.; Al Sayed, Z.R.; Hill, M.C.; Mantineo, H.; Brydon, E.M.; Zeng, Z.; Liu, X.S.; et al. Transfer learning enables predictions in network biology. *Nature* **2023**, *618*, 616–624.
8. Yang, F.; Wang, W.; Wang, F.; Fang, Y.; Tang, D.; Huang, J.; Lu, H.; Yao, J. scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data. *Nat. Mach. Intell.* **2022**, *4*, 852–861.
9. Hao, M.; Gong, J.; Zeng, X.; Liu, C.; Guo, Y.; Cheng, X.; Wang, T.; Ma, J.; Zhang, X.; Song, L. Large-scale foundation model on single-cell transcriptomics. *Nat. Methods* **2024**, *21*, 1481–1491.
10. Wen, H.; Tang, W.; Dai, X.; Ding, J.; Jin, W.; Xie, Y.; Tang, J. CellPLM: Pre-training of cell language model beyond single cells. In Proceedings of the 12th International Conference on Learning Representations (ICLR), Vienna, Austria, 7–11 May 2024.
11. Rosen, Y.; Brbic, M.; Rozenblatt-Rosen, O.; Leskovec, J. Universal cell embeddings. In NeurIPS Workshop on New Frontiers in Learning, Control, and Dynamical Systems, New Orleans, LA, USA, 15 December 2023.
12. Levine, D.; Rizvi, S.A.; Lévy, S.; Pallikuth, C.; Zhang, X.; Chen, E.; Zhang, J.; Ghadermarzi, S.; Wu, H.; Zheng, D.; Sontag, D.; Pe'er, D. Cell2Sentence: Teaching large language models the language of biology. *bioRxiv* **2024**, 2023.09.11.557287.
13. Chen, Y.; Bhatt, H.; Bhatt, A.; Liu, C.; Langdon, A.; Li, C.L.; Bhatt, P. GenePT: A simple but effective foundation model for genes and cells built from ChatGPT. *bioRxiv* **2024**, 2023.10.16.562533.
14. Radford, A.; Kim, J.W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. Learning transferable visual models from natural language supervision. In Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021; pp. 8748–8763.
15. Zhai, X.; Mustafa, B.; Kolesnikov, A.; Beyer, L. Sigmoid loss for language image pre-training. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 11975–11986.
16. Peebles, W.; Xie, S. Scalable diffusion models with transformers. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 4195–4205.
17. Lipman, Y.; Chen, R.T.Q.; Ben-Hamu, H.; Nickel, M. Flow matching for generative modeling. In Proceedings of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 1–5 May 2023.
18. Albergo, M.S.; Vanden-Eijnden, E. Building normalizing flows with stochastic interpolants. In Proceedings of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 1–5 May 2023.
19. Liu, X.; Gong, C.; Liu, Q. Flow straight and fast: Learning to generate and transfer data with rectified flow. In Proceedings of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 1–5 May 2023.
20. Stuart, T.; Butler, A.; Hoffman, P.; Hafemeister, C.; Papalexi, E.; Mauck, W.M.; Hao, Y.; Stoeckius, M.; Smibert, P.; Satija, R. Comprehensive integration of single-cell data. *Cell* **2019**, *177*, 1888–1902.
21. Brennecke, P.; Anders, S.; Kim, J.K.; Kolodziejczyk, A.A.; Zhang, X.; Prosdocimi, V.; Dinar, B.; Teichmann, S.A.; Marioni, J.C.; Heisler, M.G. Accounting for technical noise in single-cell RNA-seq experiments. *Nat. Methods* **2013**, *10*, 1093–1095.
22. Wilcoxon, F. Individual comparisons by ranking methods. *Biom. Bull.* **1945**, *1*, 80–83.

23. Franzén, O.; Gan, L.M.; Björkegren, J.L.M. PanglaoDB: A web server for exploration of mouse and human single-cell RNA sequencing data. *Database* **2019**, *2019*, baz046.
24. Hu, C.; Li, T.; Xu, Y.; Zhang, X.; Li, F.; Bai, J.; Chen, J.; Jiang, W.; Yang, K.; Ou, Q.; et al. CellMarker 2.0: An updated database of manually curated cell markers in human/mouse and web tools based on scRNA-seq data. *Nucleic Acids Res.* **2023**, *51*, D870–D876.
25. Traag, V.A.; Waltman, L.; van Eck, N.J. From Louvain to Leiden: Guaranteeing well-connected communities. *Sci. Rep.* **2019**, *9*, 5233.
26. Tabula Muris Consortium. Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris. *Nature* **2018**, *562*, 367–372.
27. Ioffe, S.; Szegedy, C. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In Proceedings of the 32nd International Conference on Machine Learning (ICML), Lille, France, 6–11 July 2015; pp. 448–456.
28. Hendrycks, D.; Gimpel, K. Gaussian error linear units (GELUs). *arXiv* **2016**, 1606.08415.
29. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* **2014**, *15*, 1929–1958.
30. Loshchilov, I.; Hutter, F. Decoupled weight decay regularization. In Proceedings of the 7th International Conference on Learning Representations (ICLR), New Orleans, LA, USA, 6–9 May 2019.
31. Loshchilov, I.; Hutter, F. SGDR: Stochastic gradient descent with warm restarts. In Proceedings of the 5th International Conference on Learning Representations (ICLR), Toulon, France, 24–26 April 2017.
32. Tong, A.; Malkin, N.; Huguet, G.; Zhang, Y.; Rector-Brooks, J.; Fatras, K.; Wolf, G.; Bengio, Y. Improving and generalizing flow-based generative models with minibatch optimal transport. *Trans. Mach. Learn. Res.* **2024**.
33. Esser, P.; Kulal, S.; Blattmann, A.; Entezari, R.; Müller, J.; Saini, H.; Levi, Y.; Lorenz, D.; Sauer, A.; Boesel, F.; et al. Scaling rectified flow transformers for high-resolution image synthesis. In Proceedings of the 41st International Conference on Machine Learning (ICML), Vienna, Austria, 21–27 July 2024.
34. Ho, J.; Salimans, T. Classifier-free diffusion guidance. In Proceedings of the NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications, Virtual, 13 December 2021.
35. Polyak, B.T.; Juditsky, A.B. Acceleration of stochastic approximation by averaging. *SIAM J. Control Optim.* **1992**, *30*, 838–855.
36. Hubert, L.; Arabie, P. Comparing partitions. *J. Classif.* **1985**, *2*, 193–218.
37. Strehl, A.; Ghosh, J. Cluster ensembles—A knowledge reuse framework for combining multiple partitions. *J. Mach. Learn. Res.* **2002**, *3*, 583–617.
38. Heusel, M.; Ramsauer, H.; Unterthiner, T.; Nessler, B.; Hochreiter, S. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 6626–6637.
39. McInnes, L.; Healy, J.; Melville, J. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv* **2018**, 1802.03426.
40. Wolf, F.A.; Angerer, P.; Theis, F.J. SCANPY: Large-scale single-cell gene expression data analysis. *Genome Biol.* **2018**, *19*, 15.
41. Luecken, M.D.; Buttner, M.; Chaichoompu, K.; Danese, A.; Interlandi, M.; Mueller, M.F.; Strobl, D.C.; Zappia, L.; Dugas, M.; Colome-Tatche, M.; Theis, F.J. Benchmarking atlas-level data integration in single-cell genomics. *Nat. Methods* **2022**, *19*, 41–50.
42. Fu, Z. CLOP-DiT: Structured-Metadata-Conditioned Single-Cell Latent Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers. *GitHub repository* **2025**. Available online: https://github.com/zeyufu/CLOP-DiT (accessed on 7 March 2026).